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Design principles for Product-Service Systems

- Using a washing machine as an example

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Design principles for Product-Service Systems: Using a washing machine as an example

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KTH Industriell teknik och management

Designprinciper för Produkt-tjänstesystem: Exemplifierat med en tvättmaskin

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Sammanfattning

Product-Service Systems (PSS) är ett växande paradigm i vilket produkter och tjänster integreras för att öka värdeerbjudandet åt användare. Detta examensarbete fokuserade på det specifika kontextet tvättmaskiner för att analysera de möjligheter och utmaningar som förknippas med att designa produkter för PSS. Forskningsmetodiken för detta examensarbete, vilket utfördes av oss på uppdrag av Electrolux AB, innefattade litteraturstudier inom bland annat området cirkulär ekonomi (CE), intervjuer med både anställda på företaget samt intervjuer och observationer av användare för att samla insikter och perspektiv från flera intressenter. Syftet med detta examensarbete var att utveckla designprinciper för PSS och presentera dessa i formen av ett koncept för en skräddarsydd tvättmaskin åt PSS.

Vår analys betonade vikten av att kunna förstå och tolka användarmönster, olika typer av användares behov, produkternas miljöpåverkan och cirkulära affärsmodeller när man ska designa produkter för PSS. De framtagna principerna betonar bland annat behovet av god reparer- och uppgraderbarhet på plats hos primäranvändare, underlättade installationer, sömlös integrering av digitala teknologier, effektiv energi- och resurspåverkan, användarcentrerad design, samt effektiva kommunikationskanaler mellan användare och tjänsteleverantörer. Både design principerna och tillika konceptet erbjuder en värdefull och överskådlig inblick åt produktdesigners, tjänstedesigners samt tillverkare som ämnar att skapa framtida PSS-lösningar. Avslutningsvis blev resultaten av detta projekt delvis relevanta för ytterligare produktkategorier och kan bidra till den bredare omställningen till PSS inom områdena produkt- och tjänstedesign.

Nyckelord: Produkt-tjänstesystem, Cirkulär ekonomi, Cirkulär design, Tvättmaskiner

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Abstract

Product-Service Systems (PSS) represents an emerging paradigm in which products and services are integrated to provide enhanced value to customers. By focusing on the specific context of a washing machine, this project explored the opportunities and challenges associated with designing for PSS. This project's research methodology, which we conducted on behalf of Electrolux AB, employed a combination of literature review, e.g. within Circular Economy (CE), interviews with company employees, as well as interviews and observations of multiple user types to gather insights and perspectives from multiple stakeholders. The purpose of this master thesis project was to develop design principles for PSS and present those in a design proposal for a washing machine bespoke for PSS.

Our findings highlight the importance of considering various factors such as user behaviour, considerations of multiple user categories, the environmental impact of products, and circular business models in designing for PSS. The developed principles emphasise the need for on-site reparable and upgradable products, easy installations, seamless integration of digital technologies, energy and resource efficiency, user-centric design, and effective communication channels between users and service providers. The developed design principles and the design proposal provide valuable guidance for designers, manufacturers, and service providers seeking to develop innovative PSS solutions. They offer a platform for creating sustainable, user-centred, and economically viable products and services that meet customers' evolving needs and preferences. The insights gained from this research can also be partly generalised to other domains and contribute to the wider adoption of PSS as a transformative approach in the fields of product and service design.

Keywords: Product-Service Systems, Product-as-a-Service, Circular economy, Circular design, Washing machines

FOREWORD

We warmly thank our KTH Royal Institute of Technology supervisor, Mia Hesselgren, for supporting our project. Furthermore, a warm thanks to our industrial supervisor from Electrolux, Adam Szczepanowski, for offering us the opportunity to undertake this project, introducing us to the company's design team, and supporting us from beginning to end. We would also like to thank all interviewees and workshop participants for taking the time to contribute to our research.

Adam Lagercrantz Zachrisson & Patrik Marusik

Stockholm, June 2023

NOMENCLATURE

The abbreviations mentioned in this report are described on this page.

Abbreviations

PaaS	Product as a Service
PSS	Product-Service Systems
CE	Circular Economy
DfD	Design for Disassembly
DfX	Design for X
R&D	Research and Development
PCB	Printed Circuit Board
UI	User Interface
CAD	Computer Aided Design
EU	European Union
WM	Washing machine
EDD	Emotionally durable design
DfSB	Design for sustainable behaviour
B2B	Business to business

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1 INTRODUCTION

This chapter describes the introduction to this project, i.e. the background, purpose, delimitations and process used.

1.1 Background

The interest in Product-Service Systems (PSS) is rising across multiple industries, and the domestic appliance industry is no exception. The potential benefits of PSS business models include boosted economic growth and more sustainable resource consumption (Reim et al., 2015). Furthermore, as a possible step in transitioning towards the circular economy, several major industry incumbents have started to pilot PSS business models in which consumers subscribe to refurbished appliances. This includes the incumbent Electrolux, which has launched a pilot Product-as-a-Service (PaaS) business model in Singapore called "Levande", where its potential is currently being investigated. In brief, PaaS can be defined as a subtype of PSS in which products are leased or rented, and the provider has ownership of them (Orasmaa, Laurila and Liimatainen, 2020).

From a sustainability perspective, the continued growth of E-waste, i.e. electrical or electronic waste, is a major concern for the environment. Most notably, large domestic appliances, including washing machines, account for more than half of the total share of E-waste in the European Union (European Parliament, 2020). According to the United Nations Global E-waste monitor report (2020), 53.6 million metric tonnes (Mt) of E-waste was generated globally in 2019, an increase of 21% during the last five years. Furthermore, the growing E-waste issue aligns with the UN Sustainable Development Goal number 12.5, which states: "By 2030, substantially reduce waste generation through prevention, reduction, recycling and reuse" (United Nations, 2015). Circular business models cater to this target by focusing on maintaining the value of goods for longer through strategies such as reuse and refurbishing.

Although PSS business models may result in more economically-, socially-, and environmentally sustainable interaction with products, the current research on applying PSS in practice is sparse (Reim et al., 2015). One of the challenges faced by Electrolux is to analyse how the design of their washing machines (WM) is affected by the change to a PaaS business model. For example, certain parts of the machine are prone to more wear and tear after they have been leased several times, which needs to be analysed in the context of a circular business model.

Lastly, the washing machines in Electrolux's product line were initially designed to be sold to consumers and, therefore, not adapted for a PaaS business model from the outset. Manufacturers often use their ordinary offerings instead of adapting them to PSS, aiming to lower the financial risk, albeit researchers suggest that adapting a product to PSS would likely be more profitable as it would lead to an increased product lifetime (Sundin, Lindahl and Ijomah, 2009).

1.2 Purpose

This master thesis project aims to analyse current washing machines in the Electrolux product line and identify challenges and important characteristics for delivering an outstanding consumer experience when offered in a foremost economically and environmentally sustainable PaaS business model. Hence, the project will identify design prerequisites needed by the design team to make a washing machine fit in a circular business model where it will be passed on to several users during its lifetime. The design prerequisites are developed as principles to be used as a guiding tool when designing for PaaS.

Furthermore, by using the principles, the second purpose is to develop a design proposal for an Electrolux washing machine bespoke for a PaaS business model. The proposal will cover one or several critical parts of the machine that are either added, removed or redesigned for circularity and enhance the consumer experience of leasing a used laundry appliance. The final result will be visualised as a design proposal for review and further development. The project comprises the following areas:

- Delivery, installation and retrieval
- Cost-effective repairability for Electrolux
- Parts harvesting
- Detergent replenishment
- Product take back
- Product refurbishment and upgrade
- Consideration of second, third or fourth users
- Aligning with the company brand values
- Consumer experience
- Sustainability

The questions to be answered that will permeate this master thesis project are related to the purpose and formulated to gain new knowledge, as follows:

- Q1: What prerequisites are appropriate for design teams when designing laundry appliances for a PaaS business model?
- Q2: How can a circular PaaS business model affect how laundry appliances are designed?

1.3 Delimitations

Several areas are relevant for us, i.e. master students from the master's programme Industrial Design Engineering at KTH Royal Insititue of Technology, to study with our background. However, due to the project duration of 20 weeks, the project was delimited and did not cover the following areas:

- Complete life cycle assessment
- B2B/industrial washing machines
- Solid Mechanics calculations

- Choice of materials
- Cost calculations
- User studies outside of Sweden

Although necessary to analyse whether a PSS business model is more sustainable than a direct sales model, this master thesis project will not investigate the environmental potential of the design proposal.

1.4 Process

The overarching process of the project follows the Double Diamond shown in figure 1. It is a common design process that implements Design Thinking, a framework developed by the Hasso Plattner Institute of Design at Stanford University in California, to solve complex creative problems with a human-centred approach (Meinel et al., 2011). It integrates stakeholders at different stages of the process to solve the problem with multidisciplinary teamwork in a non-linear, iterative way. The composition of methods in this project was carefully chosen to gain insights from different perspectives and to minimise the Hawthorne Effect and researcher bias. Research findings were created by triangulation, i.e., combining a minimum of three different research methods, to improve the credibility and accuracy of the data sampled (Stickdorn et al., 2018).

The project comprises three phases; Research, PaaS Design Principles and Implementation in Design. Also, each phase consists of a converging and diverging stage to iterate and synthesise the knowledge during the project until the final stage.

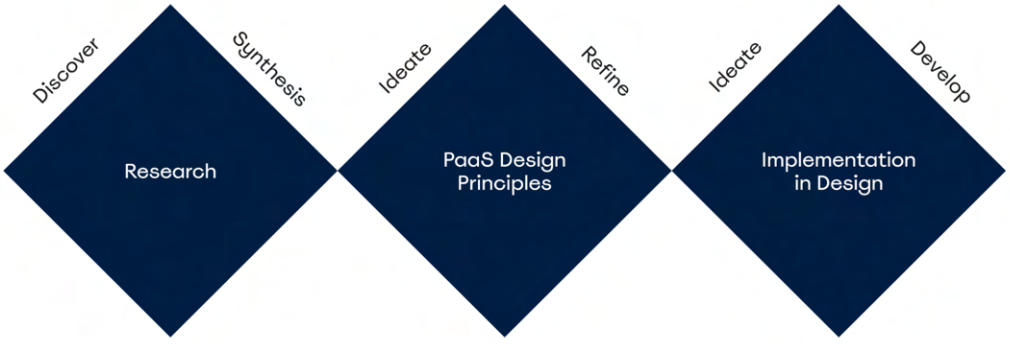


Figure 1. Visualised process, based on the Double Diamond

The frame of reference is defined for the project by reviewing literature relevant to this project. The design proposal was developed in conjunction with the client, and to ensure that the results were relevant to Electrolux, weekly meetings were set up with the supervisor during the project. The design principles were evaluated with other Electrolux Care Design Team employees through a creative workshop.

2 FRAME OF REFERENCE

This chapter contains the chosen frame of reference, which frames the context of this project in existing literature.

2.1 The context of Sweden

Since the user studies made in this project were held in Sweden, it is necessary to understand the context behind laundry practices in the country. The Swedes' laundry practices share several commonalities with other European nationalities and foremost citizens of the other Nordic nations. However, a key difference is the variety of tenure statuses which makes the nation's housing market unique and affects how laundry appliances are accessed. There are four categories of tenure status of households in Sweden, namely Right of tenancy, Tenant-ownership, Cooperative tenancy and Ownership (Boverket, n.d.). Tenant ownership is specific to Sweden, meaning that the person is a member of the tenant-ownership association, where the association owns the building, and each member has purchased the right of use to their apartment. Communal laundry rooms and home-installed washing machines are prevalent among Right of Tenancy and Tenant-ownership residents.

In fact, Electrolux was instrumental in achieving the Swedish politicians' goal from the 1950s and 60s of access to laundry facilities for all who could not afford their washing machines, which coincided with the housing boom of that period. This process began after the acquisition of "Bohus Mekaniska" in 1944, which designed and manufactured large washing machines, allowing Electrolux to enter the commercial market, see figure 2. Eventually, the communal laundry rooms were established across the country and featured Electrolux machines branded under the name Bohus (Electrolux Group, n.d.). The communal laundry room is still a prominent part of Swedish living and free for all residents in the apartment building, which is one of the reasons why laundromats have not gained mainstream adoption in Sweden.

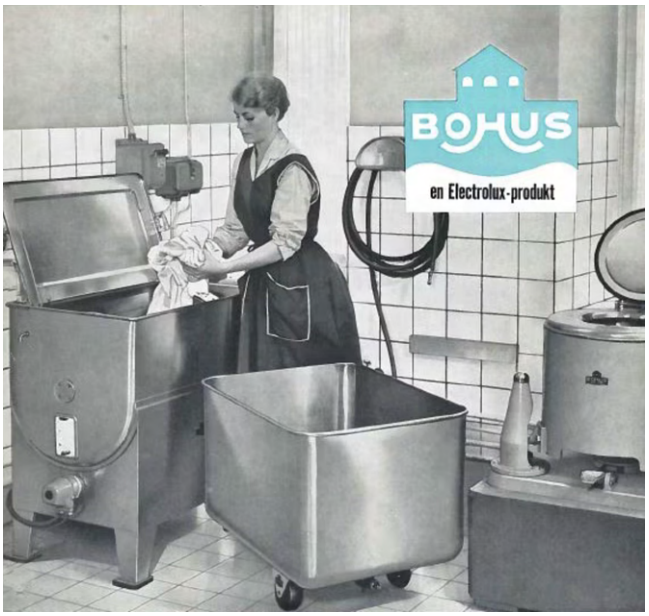


Figure 2. Electrolux Bohus, a washing machine in communal laundry rooms (Electrolux Group, n.d.)

Another essential aspect to note is that Swedes often leave their appliances behind when they move out, including washing machines and dryers. This is mandated in the case of tenant ownership, where the association has supplied the appliances since they own them (Mäklarringen, n.d.). Hence, laundry appliances are often inherited from previous residents. That said, many Swedes live in private houses and might have larger freedom in choosing their major domestic appliances. In regard to specific laundry routines in Sweden, according to a study made by Electrolux (Dagens Nyheter, 2021), 40 degrees is the most common washing temperature and powder detergents are mainly used over other kinds among Swedes, despite their reduced effectiveness at lower temperatures. Furthermore, statistics from Eurostat (European Union, n.d.) show that Sweden is the country in the EU where its adolescents move out from their parental home the earliest, on average at 19,0 years old. For reference, the highest number is Portugal at 33,6. Hence, Swedish adolescents are learning to live independently, including performing household chores, from a relatively early stage.

Lastly, regarding Swedes' current perception of subscription services for washing machines, previous studies have indicated a general interest in renting them instead of direct sales (Lieder et al., 2018), which affirms the context of this master thesis project, to an extent. Lieber et al. mention that including additional services is the most contributing factor for securing the consumers' interest and that the respondents are mainly willing to subscribe to a lease of washing machines based on reduced CO2 footprint, provided that the potential environmental gain is expressed clearly. Note that these findings are based on data from Stockholm and may not reflect the opinions on a national scale.

2.2 Front-loaded washing machines

The basic structure of a front-loaded washing machine consists of a drum, a drum paddle, a motor, a pump, and a water inlet and outlet. The drum is the part of the machine where the clothes are placed, and it rotates during the wash cycle. The drum paddle moves the clothes around in the drum, ensuring they are evenly cleaned.

The washing process begins by filling the drum with water. The water is supplied through the water inlet valve, controlled by the washing machine's electronic control board. Once the drum is filled with water, it begins to rotate and adds the detergent. The rotation then moves the clothes around in the drum, allowing the detergent to penetrate the fabric. The drum then rotates, creating a scrubbing motion that helps to remove dirt and stains from the clothes. The water is then drained from the drum, and freshwater is added for the rinse cycle. During the rinse cycle, the drum rotates again, and the water is drained from the machine again. Once the rinse cycle is complete, the machine enters the spin cycle. During the spin cycle, the drum rotates at high speed, which helps to remove excess water from the clothes. The water is pumped out of the machine, and the clothes are left damp and ready to dry.

Moreover, all washing machines sold in the EU must declare their energy consumption (European Commission, n.d.). Figure 3 shows an excerpt from an Electrolux manual.

Dessa värden är endast indikerande.

Program	kg	kWh	Liter	tim:min	%	°C	varv/min[1)]
Bomull[2)] 90 °C	9	2.15	85	3:35	44	85	1600
Bomull 60 °C	9	1.55	80	3:50	44	55	1600
Bomull[3)] 20 °C	9	0.35	80	3:10	44	20	1600
Syntet 40 °C	4	0.80	50	2:25	35	40	1200
Fintvätt[4)] 30 °C	2	0.30	45	1:10	35	30	1200
Ylle 30 °C	2	0.25	65	1:05	30	30	1200

1) Referensindikator på centrifughastighet.
2) Lämpligt för mycket smutsig tvätt.
3) Lämplig för lätt smutsad tvätt av bomull, syntet och blandade material.
4) Det fungerar också som ett snabbt tvättprogram för lätt smutsad tvätt.

Figure 3. Energy consumption for an Electrolux washing machine (Electrolux Sverige, n.d.)

2.3 Current PaaS business models at Electrolux

Levande

Electrolux has started to pilot a PaaS business model in Singapore called “Levande”, where the consumer leases domestic appliances by subscribing to the service for a duration of six months to six years (Levande, n.d.). The consumer may choose between either a new or refurbished product when ordering. The monthly fee depends on the product's condition and the subscription's length. The service includes free maintenance, repairs and like-for-like replacements during the subscription. The consumer can choose between four washing machines, including three front-loaded and one top-loaded variant (ibid.). The washing machines are off-the-shelf products that can be purchased through regular sales channels and have not been adapted explicitly for PaaS. Among the goals for Levande is to expand to other markets.

Figure 4 shows the high-level circular flow of Levande, with each step in the service detailed. As shown in the figure, the machine will always pass through the Electrolux warehouse after the end of each subscription. This stage involves an extensive cleaning process in order to prepare the WMs for the next lease cycle. The biggest hurdle in this process is cleaning the drum, which requires removing the main component from the machine.

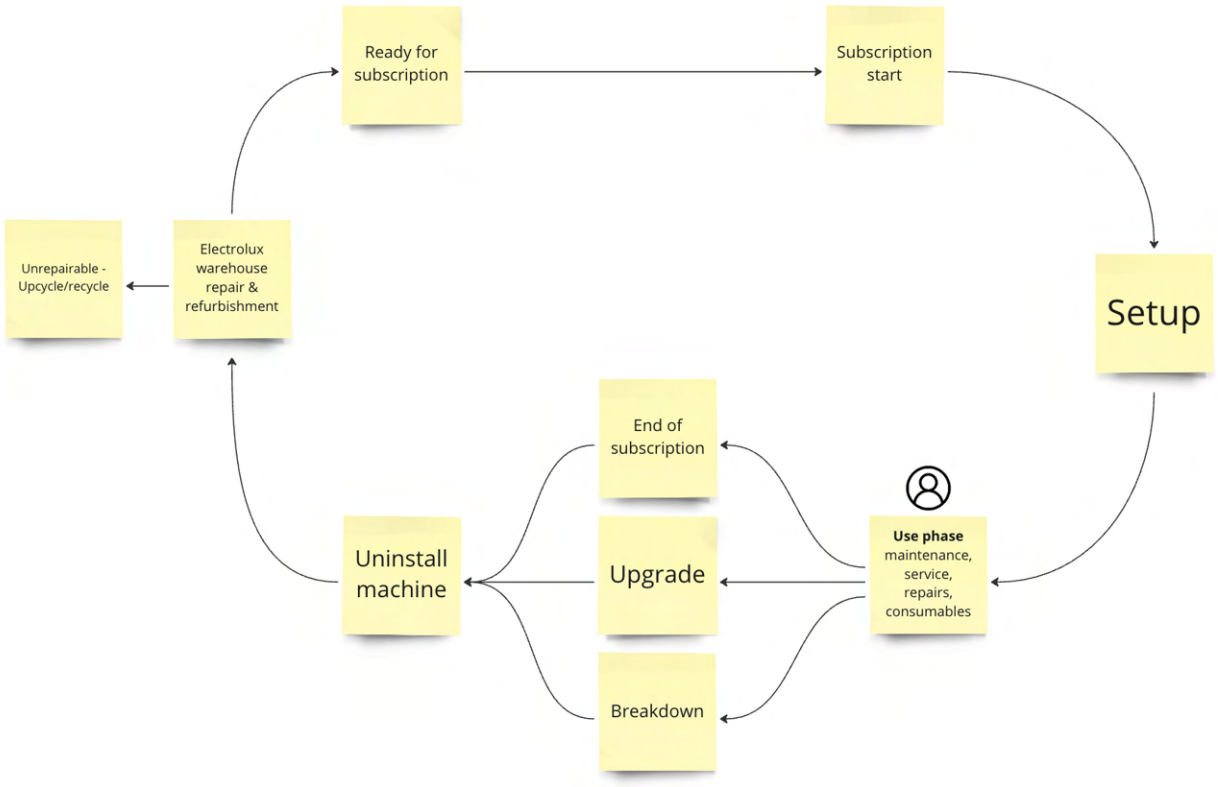


Figure 4. The high-level circular flow of the service Levande

Pure i9.2 robot vacuum cleaner

The second PaaS model from Electrolux is the Pay-per-use business model offered in Sweden with their Pure i9.2 robot vacuum cleaner. The consumer is charged monthly according to the amount of area the vacuum has cleaned. Servicing is included, and after 4000 m² of cleaning, a set of new brushes and filters are sent to the consumer at no additional cost. If the vacuum breaks down, Electrolux will repair it or send a replacement at no additional fee. Moreover, the service requires no deposit, and the consumer may end the subscription at any time (Electrolux, n.d.).

2.4 Circular economy

According to The Ellen MacArthur Foundation (2015), the term Circular Economy is characterised as an economy which is both "generative and restorative" to maximise value and utility at all phases of the product life cycle. Moreover, CE is built on three systemic principles, namely, "Preserve and enhance natural capital", "Optimise resource yields", and "Foster system effectiveness". These principles are visualised in figure 5, referred to as "The Butterfly Diagram", which maps the entire system of CE. The diagram distinguishes between technological and biological cycles. The latter emphasises the processing of natural resources back into the earth, while the former consists of several strategies for prolonging product lifetime, e.g. reuse, refurbish or remanufacture. Recycling, refurbishment and maintenance are the strategies which the current service targets. As shown in the figure, this forms a circular bond between the user and the product manufacturer, in this case, Electrolux. The principal goal of the technological cycle, highlighted in blue, is to prolong the lifetime of goods and maintain their value for as long as possible. This can be achieved by keeping the product in the innermost loops without exchanging parts for as long as possible.

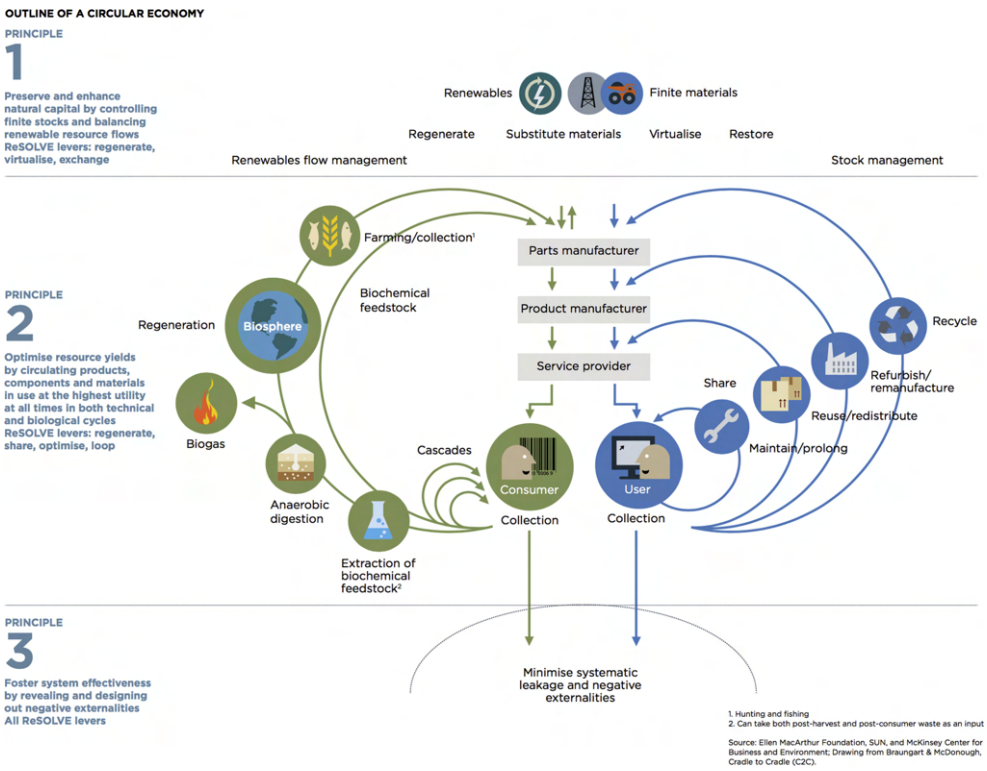


Figure 5. The Butterfly Diagram of Circular Economy (Ellen MacArthur Foundation, n.d.)

Although the concept of CE has existed for over a decade, the research for incorporating it into design has mainly been focused on existing fields by an interdisciplinary approach using Design for X (DfX) strategies. The variable X can be exchanged for several methodologies to address specific issues during a product's life cycle. Most methods are limited to the manufacturing, remanufacturing and recycling aspects, mainly requiring skills from various engineering fields. Studies show that addressing circular design issues requires transdisciplinary skills, not only from engineering fields but also competencies from human practices and social behaviours (De los Rios and Charnley, 2017). The latter are essential skills for understanding the users' roles in a closed-loop system and enhancing the consumer experience and are mainly those that the design department at Electrolux possesses. This implies that the use phase involving the Share/Maintenance/Prolong strategies should be given the most attention in this project to aid the Electrolux design department with developing products for PaaS.

2.5 Design for Circularity

Design has a crucial role in shaping the circular economy, which has given birth to the concept of Circular Design. A fully circular design can be thought of as a design which has transitioned from the linear thinking of manufacture-use-disposal to a situation where the end-of-life phase has been eradicated and continues to live on through new cycles of life. Although several frameworks exist for DfX strategies, e.g. Design for remanufacture, few focus on product design in the context of CE.

The Delegation for Circular Economy (2020) has devised a framework consisting of ten Circular Design strategies based on the three CE principles mentioned in the previous chapter (Ellen MacArthur Foundation, 2015), which details the characteristics which make either products, services or business models suited to Circular Economy. Apart from distinguishing between the different CE principles, the strategies are further divided into before, during or after use, see figure 6.

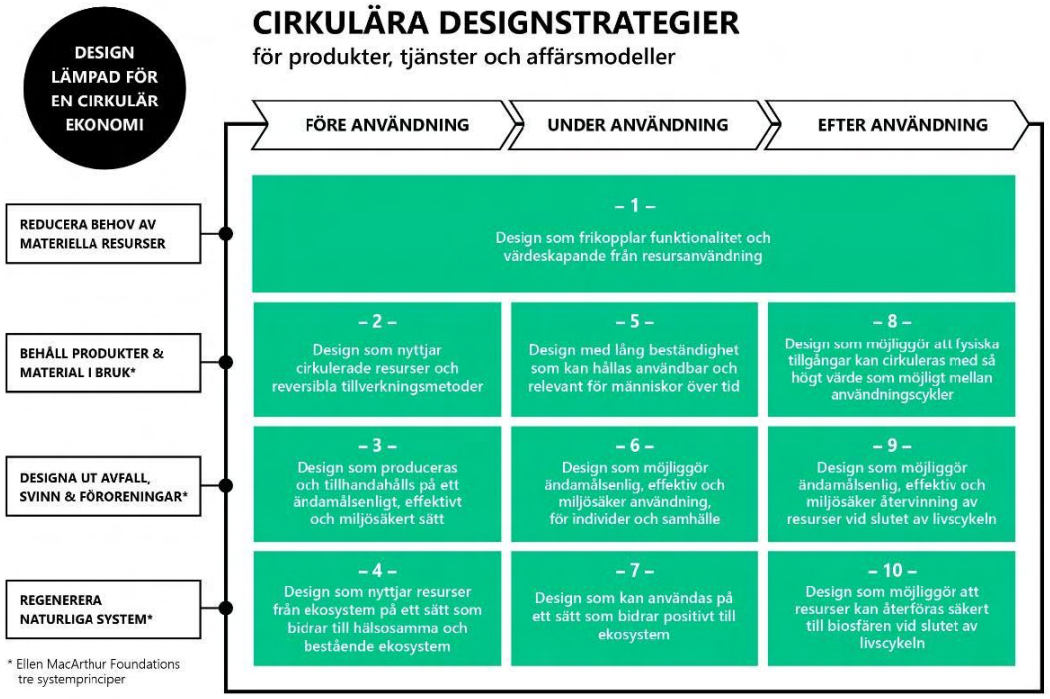


Figure 6. Ten Circular Design strategies (The Delegation for Circular Economy, 2020)

Regarding the service of Levande, in its current state, the business model mostly pertains to strategy eight regarding refurbishing the appliances after the leasing contracts expire or are otherwise cancelled. This master thesis project seeks to find ways of applying more strategies from the preceding life cycle phase, i.e. during use. Strategies focused on prolonging product life are interesting since this plays a significant role in achieving product circularity. Referencing Strategy five, the design proposal should be designed for longevity, durability, and relevance for the consumer long-term. This includes both usefulness and emotional attachment. An example of strategy five is Skeppshult's cast iron skillets which improve their performance over time, and can last for generations (Skeppshult, 2017).

Subsequently, strategy six involves a design that enables appropriate, efficient and environmentally safe usage for individuals and society. Finally, Strategy seven calls for a design that can be used to contribute to an ecosystem positively.

Another product design framework for CE, to aid product designers by adapting the CE principles to applicable design guidelines, has been proposed by researchers from TU Delft (Van den Berg and Bakker, 2015). The framework consists of three tools, where the first gives a brief

outline of what topics should be addressed in the circular design project, see figure 7. The main topics are; Future proof, Disassembly, Maintenance, Remake and Recycle. Why they are needed and how they correlate is described in the figure.



Figure 7. The areas that should be addressed in a circular design project (Van den Berg and Bakker, 2015)

The second tool is a set of guidelines where the goals for the topics are stated together with the means needed to achieve them, the overview can be seen in figure 8, and the complete guidelines can be seen in Appendix A.

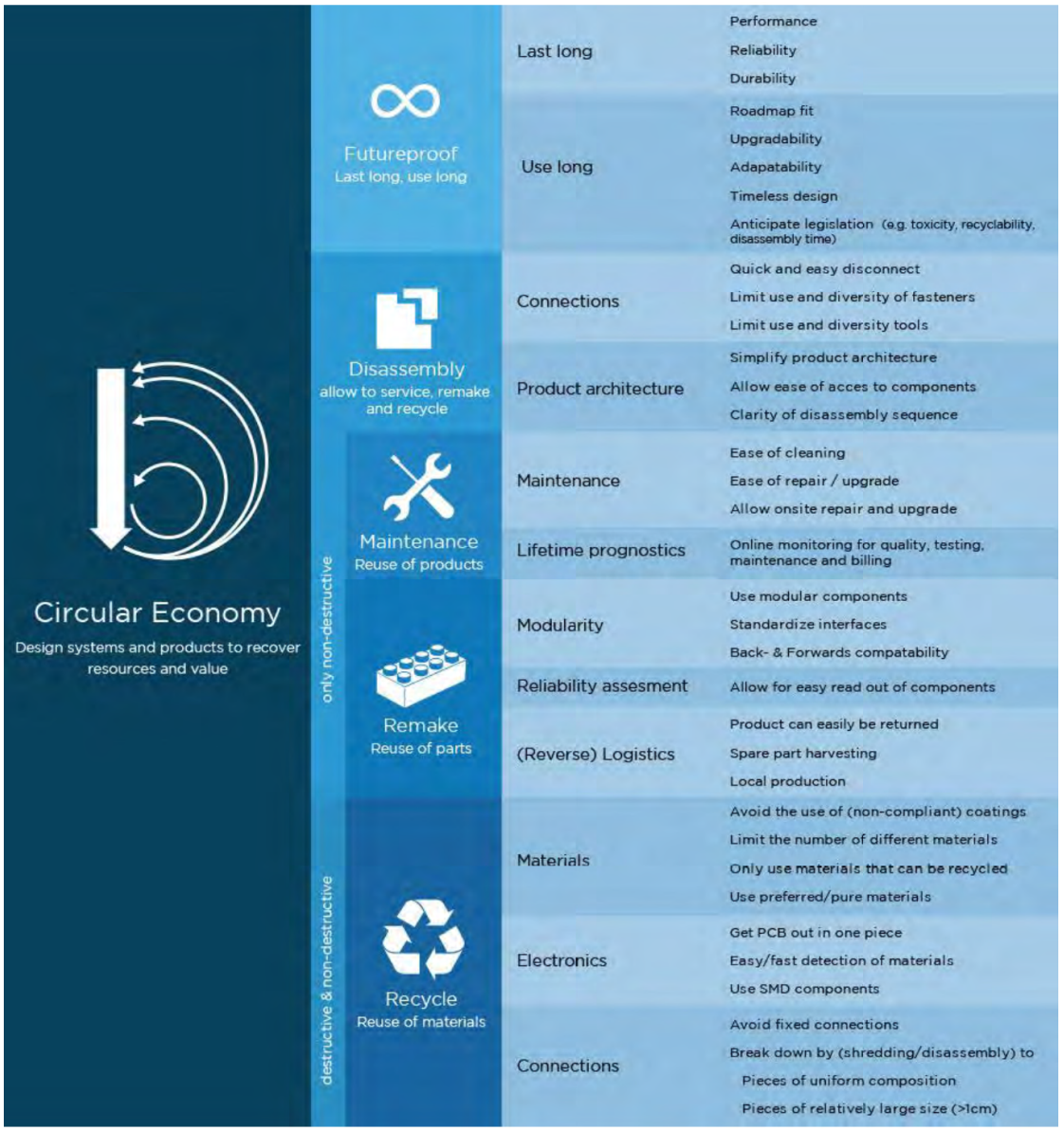


Figure 8. CE design guidelines proposed by researchers from TU Delft (Van den Berg and Bakker, 2015)

The guidelines are extensive, with referrals to comprehensive design approaches and a mixture of firm requirements with softer, subjectively stated considerations. Some guidelines involve mechanical dimensioning and material properties, requiring a background in mechanical engineering or similar. These guidelines mainly consider the manufacturer's perspective and assume that the service provider takes care of the machine rather than putting the responsibility on the consumer. Note that the consumer's role is not in focus.

The last tool of the framework consists of a spider map for evaluating and comparing circular design concepts with regard to the topics, see figure 9. Important to note is that the spider map does not weigh the importance of the different topics, although certain are more likely to be more important than others, as shown in figure 5.

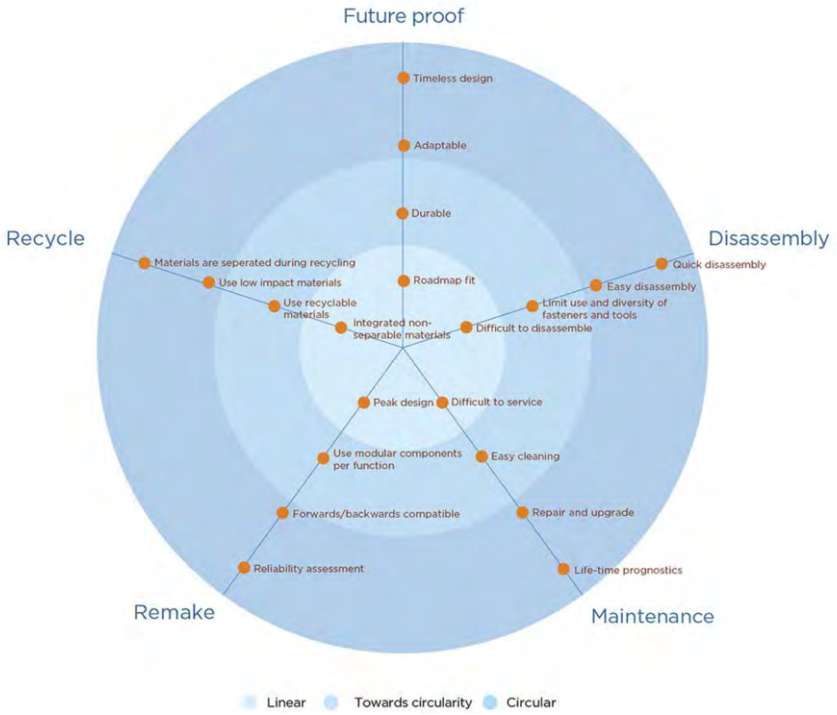


Figure 9. A spider chart for circular design evaluation (Van den Berg and Bakker, 2015)

Van den Berg and Bakker (2015) concluded that the guidelines could be used for the detailed design, and their validation showed that participants mainly used it after the ideation phase, opting for further development of an approach that facilitates the work in the earlier project phases.

It can be explored whether the presented design for circularity frameworks is appropriate for use by the Electrolux design department. However, the frameworks imply that competencies from several different fields should work multidisciplinary to develop circular products. Therefore, guidelines tailored for the design department's skills to be used as prerequisites for designing for PaaS should be investigated further.

2.6 Product-Service Systems

Product-service Systems (PSS) are business models that fulfil specific customer needs by combining tangible products with intangible services in a joint system (Tukker, 2004). Furthermore, Tukker recognises three main categories of PSS, each depending on the ratio of product content versus service content. First, in Product oriented PSS, the customer purchases the product and has ownership. However, the provider may accrue additional revenue by selling service and maintenance packages. Second, in Use oriented PSS, the provider retains ownership and may offer the said product through, e.g. leasing, sharing or renting. Third, Result oriented

PSS entails the customer and provider agreeing on a result, and the means for achieving it are left open to the provider. Moreover, the main categories consist of eight specific architectural subcategories with different characteristics, see figure 10.

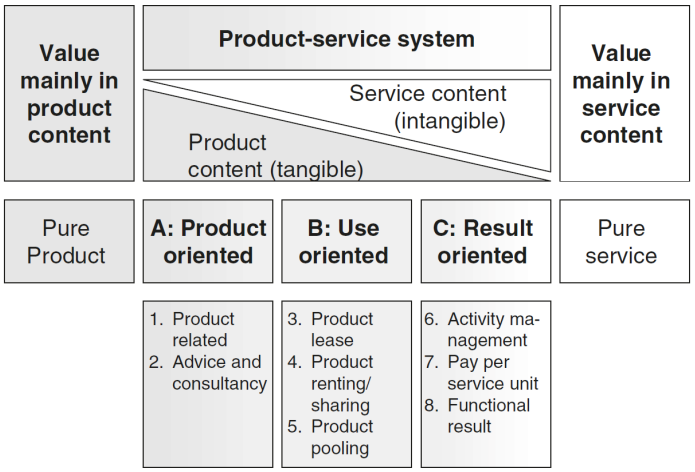


Figure 10. The Product-service system models

Electrolux service Levande owns the machine during the lease and is responsible for repairs and maintenance; hence it can be identified as a Use oriented Product lease model. When moving from a pure product to Product leasing, the tangible consumer-added value increases as some of the costs and tasks shift from the user to the provider, thus also increasing the initial financial investment needed for the service. On the other hand, since the investment for the consumer will be lower than purchasing the product, it can be easier to attract new customers and provide, e.g., maintenance and repairs could increase their loyalty. Since the customer does not own the product, there might also be a risk of them using the machine more carelessly, and if they do not feel loyalty to the service, it can also be easy for them to change to another provider (ibid.). The sustainability potential for environmental impact reductions for this model is limited due to a low incentive for the provider to innovate for, e.g., higher efficiency during the use since the cost for consumables is not considered when the business model is designed.

Furthermore, the other two subcategories of Use oriented, namely Product renting/sharing and Product pooling, both come with their advantages and disadvantages. Both have a high environmental potential and Product pooling, especially since the products are used by multiple people concurrently. Granted, as in relation to Product lease, their environmental gain necessitates being expelled from potential product misuse. Furthermore, in the case of Product renting/sharing, it is imperative that the Use phase does not dominate in regard to environmental impact.

Regarding Product oriented PSS, the two types are ‘Product related’ and ‘Advice and consultancy’. The latter means that in addition to selling the product, the provider also offers advice on how to operate it best. Whereas the former means that the provider also offers, for

example, a maintenance package during the use phase of the item. These are not particularly challenging to implement for established firms, albeit the potential environmental gain is low.

Subsequently, Result oriented PSS consists of Activity management, Pay per service unit, and Functional result. First, in Activity management PSS, all personnel and material costs are absorbed by the provider, and certain activities within the company are outsourced to third parties. Second, in Pay per service unit, the customer is only charged according to how much he/she uses the product. Lastly is Functional result PSS, where the provider only negotiates with the customer on an end result and is not bound by a specific technology. According to Tukker (2004), Functional result has the potential for the highest environmental gain of all PSS models since the means are left fully open, and the provider can ensure that the target is being met in the most efficient manner possible. Conversely, this category is associated with the highest risk and level of uncertainty.

Note that PaaS is absent from these categories. According to Orasmaa, Laurila and Liimatainen (2020), PaaS is defined as either renting or leasing of products in which the service provider owns the product. Thus, the definition corresponds the most with Tukker's (2004) notion of a Use-oriented Product lease. To ensure that the new PaaS business model is sustainable and profitable, the product design should strive to reduce the financial risk of the provider, increase its lifetime within the service, stay at the consumer as long as possible, promote careful usage and allow for business models with higher sustainability potential to be used.

Lastly, it is common for manufacturers to underestimate the process of transitioning from product content to a service content-based business model, and more aspects of the value chain need to be considered (Loon, Van Wassenhove and Mihelic, 2021). Highly overlooked aspects of the new business model include logistics, remanufacturing, service personnel and technicians, potential dealers, consumer relationships and adaptation to their changing needs. Therefore, it is important to use a holistic system approach when designing for PaaS.

2.7 Fostering caring behaviour of products

Extending products' lifespan is essential in transitioning to Circular Economy and service-based business models. However, increasing products' durability comes with inherent uncertainties if the product is no longer considered useful to the consumer, leading to product obsolescence (Asif et al., 2021). While it concerns ensuring that the products maintain their technological and functional value long-term, it also encompasses establishing an emotional bond with them (Ceschin and Gaziulosoy, 2020). Technological obsolescence means that products are surpassed by and rendered incompatible with new technologies, whilst functional obsolescence means that products decrease their performance over time or stop functioning altogether. While other fields of research, such as Eco-design, focus on prolonging the life of products in relation to, e.g. manufacturing, it lacks the user's perspective of why products are rendered obsolete for reasons beyond losing their functionality (ibid.).

This has given birth to the research field of Emotionally durable design (Chapman, 2015), which seeks to strengthen the emotional bond between users and products to avoid premature replacement of products and ensure that goods continue to provide value and meet the needs of the users' long-term. Ceschin and Gaziulosoy (2020) mention several design strategies for strengthening the emotional bond to products, e.g. “Designing products that age with dignity” and “Designing adaptable products”. The former is akin to the Japanese philosophy of “Kintsugi” (Sho, 2021), in which imperfections are embraced and cherished; imagine leather fabrics which improve their appearance after a long time. The latter means designing objects which adapt to new customer needs as life goes on, such as modular products. Ceschin and Gaziulosoy (2020) also mention the principle of “Self-expression: the ability of the product to express one’s unique identity”; namely, users often attach a stronger emotional bond to products with a personal identity.

As mentioned, whereas certain designs for sustainability approaches focus on reducing the environmental impact of products during their lifecycle from a technological perspective, others seek to understand the effect user behaviour with products has on their environmental performance. Design for sustainable behaviour (DfSB) is an approach that uses design strategies and techniques to encourage and support individuals in adopting sustainable behaviours and innovations (ibid.). DfSB seeks to bridge the gap between individuals’ intentions to act sustainably and their actual behaviours. It acknowledges that simply providing information or raising awareness about sustainability issues may not be enough to drive behavioural change. Instead, it focuses on creating interventions that make sustainable behaviours more attractive, convenient, and socially desirable. However, since DfSB is still a fairly young area of study, it lacks a standardised model to explain the features of the approach (ibid.). Nonetheless, scholars within the field agree that the approach can be condensed into four principles. First, desired behaviour should be made easier to adopt. Second, unwanted behaviour should be made more challenging to execute. Third, increase users’ willingness to perform wishful behaviour. Conversely, the last is making users refrain from unwanted behaviour.

There are several examples of DfSB across various domains. For instance, the digital instrument cluster found on the first generation of the electric vehicle Nissan LEAF (Nissan USA, n.d.). It featured a graphical image of a tree, and if the driver drives eco-friendly, more tree segments are shown and vice versa.

3 METHODS

The chapter presents the methods used in this project, their purpose and how we implemented them.

3.1 Stakeholder map

The method entails constructing a visual map identifying relevant actors to the project and understanding how they relate to each other (Stickdorn et al., 2018). The purpose was to determine which actors should be prioritised to answer the project questions and how the involved parties should be managed.

Implementation

We, i.e. master students from the master's programme Industrial Design Engineering at KTH Royal Insititue of Technology, established a stakeholder map for the actors that impacted the consumer experience or could be affected by future product changes. The external actors' interest and influence on the project were subjectively determined in a chart, whereby the ones with the most influence and power should be prioritised. Also, to understand how the design at Electrolux would be affected by adapting to PaaS, we constructed a second internal stakeholder map solely for them.

3.2 Brand analysis

The eight pillars is a method of analysing brands according to eight areas to understand the ambition and perception of the brand (Hesselgren, 2022), see figure 11. The purpose was to understand Electrolux's brand identity so that the design proposal could be aligned with it.

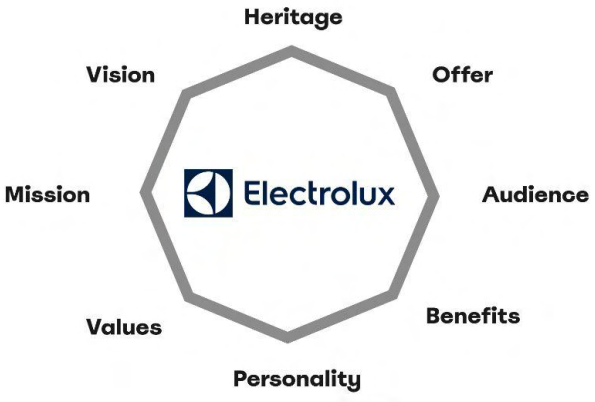


Figure 11. The eight pillars of a brand (Hesselgren, n.d.) with the Electrolux logo

Implementation

We analysed each area step by step based on the information given by the company, thus with an inward perspective and based on our subjective perception of the brand, i.e. an outward perspective.

3.3 Competitor analysis

Competitor analysis is a method for analysing the competitors' products and services to evaluate and compare them to a company's offering (Milton and Rodgers, 2013). Furthermore, the method can contribute to both functional requirements and subjective user needs, which guide the product development process. The purpose was to analyse the products on offer and their features, the additional services, and the conditions of the subscription contracts.

Implementation

We performed analyses of competing services to Levande in the European market as part of the research phase. Note that only services that partner with a major manufacturer were analysed.

3.4 In-depth interviews

In-depth interviews are used to gain insights about users' perspectives, emotions and expectations related to the research questions based on a semi-structured interview guide regarding the theme. The reason is to structure the interview to an extent and let the respondent share their thoughts openly to allow tacit knowledge to be captured (Stickdorn et al., 2018).

Implementation of interviews with Electrolux

We conducted in-depth interviews with people employed by Electrolux to understand the organisation better, the current design and product development processes, sustainability, and the role of PaaS at present and in the future. The interviews were held in a semi-structured format to retain structure whilst allowing the interviewee to express themselves freely. The interviews persisted with the most obvious questions related to the project, albeit slightly modified depending on the person's expertise. The full interview guide is shown in Appendix B. Examples of the consistently asked questions were as follows:

- What is the function of your department, and what are your purposes?
- In what ways do you think the transition towards Circular Economy will affect your role?
- In what ways do you think the transition towards Product as a Service will affect your role?
- In what ways may our project contribute to your work?

Implementation of interviews with primary users

We also held semi-structured in-depth interviews with consumers to understand the role of the primary users in a PaaS model, their attitude towards products they own versus lease, and what

the Electrolux design department needs to consider to prolong the lifetime of washing machines. The interview guide can be seen in Appendix C. Some examples of the questions asked were:

- How do you take care of washing machines?
- Do you feel you have enough knowledge regarding how to take care of washing machines?
- What would you do if your machine broke down?
- How do you relate to products that you rent versus your own?

3.5 Contextual interviews

Contextual interviews are used in a context relevant to the research, where the interviewees' surroundings and behaviour can be observed to gain insights regarding their needs (Stickdorn et al., 2018).

Implementation

We conducted semi-structured interviews in a contextual setting with consumers to understand their interaction with washing machines whilst observing them in the relevant environment. The interview guide in Appendix C served as the base for the interviews. In addition, the interviewees were asked to demonstrate how they perform laundry and common maintenance.

3.6 Customer journey map

Customer journey map (CJM) is a method for gaining insight into success factors and problem areas where design might be directed (Stickdorn et al., 2018).

Implementation

We devised a CJM of the primary users to identify pain points before, during, and after performing laundry and engaging with WMs. The purpose was to find pains, gains and opportunities for the design proposal based on the experiences of current products. We developed the CJM based on the collective insights from consumer interviews. The interactions with washing machines were analysed step-by-step, and apart from the pains, gains and opportunities, an emotional curve was constructed to reflect the consumer’s feelings throughout the journey.

3.7 Product autopsy

Product autopsy is a method that entails disassembling the products’ of interest to generate knowledge about how they are designed and engineered so that they can be developed further (Milton and Rodgers, 2013).

Implementation

We performed autopsies of existing and former configurations to understand how the design has changed from older generations. The purpose was to understand the repairability of Electrolux’s WMs, the everyday work of technicians, and the procedures in place for service call visits.

3.8 Participant observation

Participant observation is a method for understanding more profound user needs that can later be used for product design. It entails shifting between passively observing and asking questions to the observed user (Stickdorn et al., 2018).

Implementation

We performed a participant observation of an installer by observing their work routine for one day. The journeys of the installations were analysed step by step, starting with unloading the machines from the vehicle, subsequently transportation from there to the accommodation, unpackaging and set-up, and finally, possible take-back of the old machines. Foremost, the method sought to provide insights regarding the transportation of the machines, their set-up process, and the consumer interaction of the installer.

3.9 Analysis on the wall

The method is used to get a visual overview of the research insights to analyse, and patterns synthesise the data into themes which guide the design work. An added benefit of the method is providing information and inspiration for creativity (Stickdorn et al., 2018).

Implementation

We wrote all of the takeaways from the research data on notes and placed them on a digital wall in Miro (n.d.) to find patterns. These were later clustered, summarised and arranged into categories based on recurring patterns and consistent themes. Uncategorised notes were placed on a “parking lot” and returned later to be categorised. The source was written on each note to document where the data or information came from.

3.10 Brainstorming

Brainstorming is an ideation method used to generate large sets of ideas quickly and non-judgingly (Online Companion, 2019).

Implementation

We conducted the brainstorming sessions in timed intervals and structured them around the categories of insights gathered from the research. The first loop of brainstorming aimed to investigate how the summarised insights could be used as prerequisites for designing a washing machine and provide a basis to refine the summarised insights and categories to fit the project’s purpose and make them more relevant for designers. While the second loop of brainstorming was performed to generate ideas for the PaaS design proposal.

3.11 Developing key insights

Developing key insights is a method that is used to help researchers communicate their research findings concisely and actionable so that they can be used further down the design process (Stickdorn et al., 2018). They serve as a foundation for ideation sessions and can later be used to evaluate ideas, prototypes and concepts.

The key insights should be formulated carefully and specifically with a clear connection to the research so that they lead to opportunity spaces that offer diverse and relevant solutions (Online Companion, 2019). Co-creative workshops can be used as a strategy to ensure that the key insights are meaningful and useful for ideation.

Implementation

We developed the key insights by removing redundancy, condensing the insights from the analysis on the wall and prioritising which ones were relevant to create an opportunity space that yields solutions focused on giving an outstanding consumer experience. We formulated the key insights and evaluated and refined them into design principles that were meaningful, understandable and actionable.

3.12 Creative workshop

Creative workshop is a method which entails a group of users to gain insights, discuss problems and find solutions together through co-creation (Milton and Rodgers, 2013).

Implementation

We conducted the workshop with the client with competencies from different fields. The purpose was to evaluate the design principles with the participants by ideating based on them to ensure they were useful.

4 RESEARCH RESULTS

The chapter presents the results of the research methods detailed in the previous chapter.

4.1 Stakeholder map

The result for the external stakeholder map can be seen in figure 12.

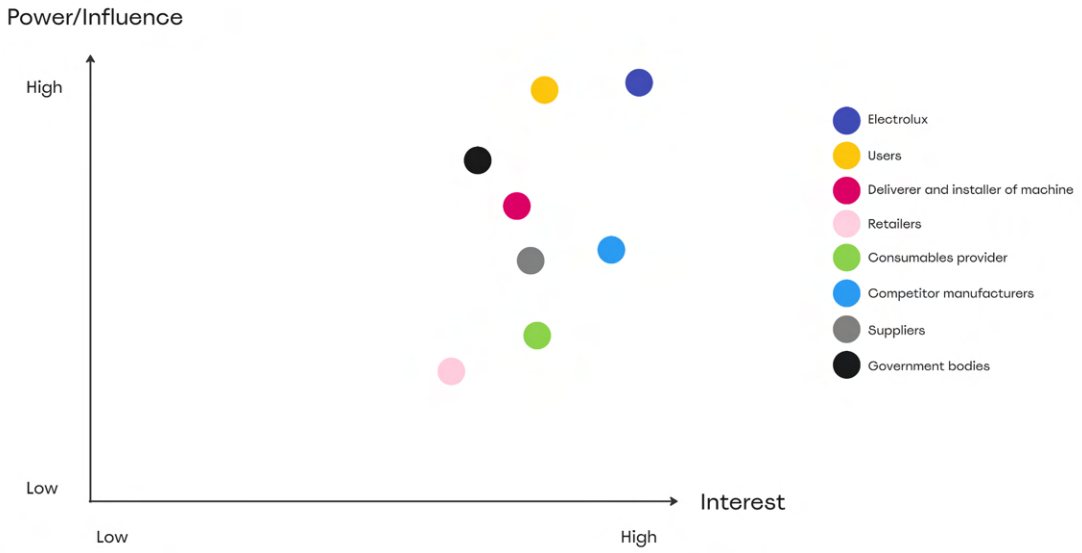


Figure 12. The stakeholder map of the different actors

First was Electrolux, rated on top in terms of Power/Influence. Their interest was also the highest since they are the leading actor in the PaaS system. Second, the consumers strongly influence Electrolux, as illustrated by the company's devotion to consumer insights and design research. The consumers' interest in PaaS business models was estimated to be fairly strong but requires further investigation. Third, the deliverer and installers' interest and Power/influence was medium to high since PaaS places higher demands on efficient transportation and swift installation and take-back of appliances. Subsequently, the retailers had some interest in the PaaS system to investigate its future potential as it could potentially reduce their sales of regular washing machines. However, since the sustainability benefits of a PaaS model mainly occur when the incentive to perform product improvements was on the manufacturer of the machines, the products should be supplied by them, leading to a low Power/influence on the design by the retailers.

Next, the consumables provider had a high interest as a PaaS model can include consumables such as, e.g. a detergent or other cleaning products, although they had moderate Power/influence on the design as their products are standardised on the market. Furthermore, competitor manufacturers are highly interested in future designs to ensure their offering stays relevant. However, they do not have as much Power/influence in the design unless they invent a disruptive

innovation. Therefore, it was also important to consider them during the design process. The suppliers' interests were moderately high since their operations would be impacted by changes to the design, although it is largely the provider's decision to make and therefore the suppliers have to accept the possible new conditions if they want to stay contracted. Lastly, government bodies had a medium to high Power/Influence. Although government bodies may legislate requirements on PaaS items, manufacturers have the freedom to design products to their liking by and large.

The second stakeholder map was constructed for Electrolux, which was classified as the actor with the highest priority in the external stakeholder map. The result can be seen in figure 13.

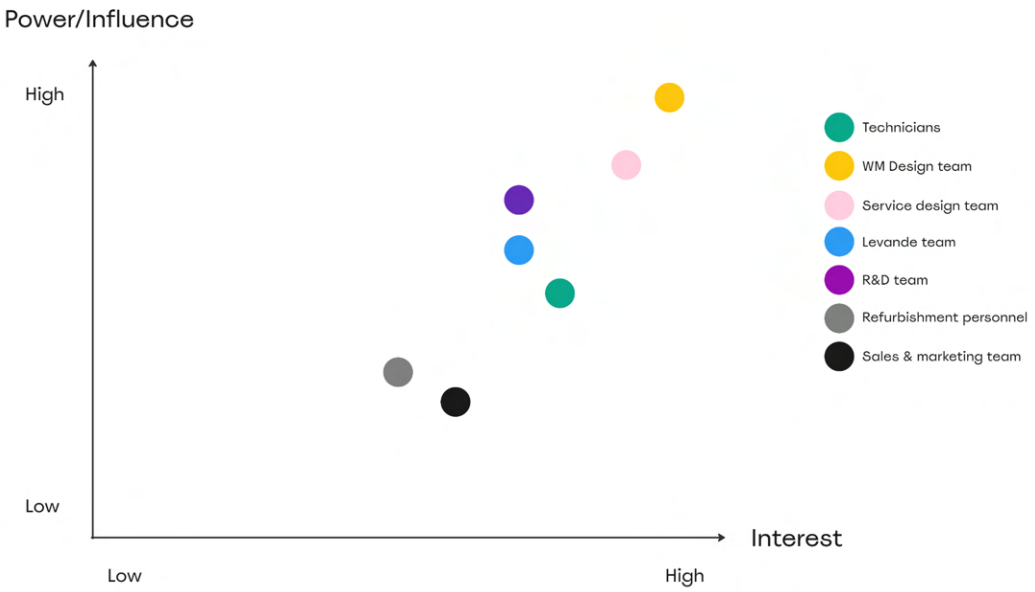


Figure 13. The stakeholder map of the internal actors in Electrolux

The WM design team was the department which took precedence in this master thesis project, and they are responsible for the Product design, which will be affected by transitioning to a PaaS model. Therefore, they were determined to have the highest influence and interest. Secondly was the Service design team, which had a strong Power/influence and interest in products that were in a Product-Service system as it became part of the service. While the Design team at Electrolux mainly focused on the user-facing areas affecting the consumer experience, the Research and Development (R&D) team focused on engineering aspects such as dimensioning, advanced development and manufacturing of the product. This means they had a medium to high Power/influence and interest in the product design since the user-facing areas must concur with the technology used.

Levande were highly interested in the project to learn and improve their current business model. The learnings from their pilot may affect the product design. However, since they operate in a different continent with cultural differences from this project's context, their Power/influence was determined to be medium to high.

Electrolux had yet to establish operations for refurbishment personnel, consequently having low Power/influence in this project. However, since they would become a part of the PSS system, they should not have been neglected. Furthermore, it could be possible that some of the current operations of the company adopt the refurbishment task, hence they were assumed to have a medium interest. Lastly, the Sales and marketing team's interests were not rated to be medium, as new sales channels and product market approaches might have become topical. Therefore, their Power/influence ratings were low compared to those mentioned earlier and, therefore, not given the highest priority in this project.

4.2 Brand analysis

The Electrolux brand was analysed to understand how the design proposal can align with the company's values.

Heritage

Electrolux was founded in Sweden in 1919 by entrepreneur Axel Wenner-Gren after the merger of companies Elektromekaniska AB and AB Lux. Electrolux stands on a long tradition of reinventing what great taste, care and well-being experiences mean for their customers, with the purpose of shaping living for the better (Electrolux Group, n.d.).

Offer

As one of the world's leading manufacturers of major and small appliances, Electrolux offers a wide variety of products ranging from coffee machines, vacuums, stoves, washers and more. Electrolux seeks to improve constantly and is synonymous with innovative, sustainable products which shape living for the better.

Audience

Electrolux's audience seeks quality products that integrate fully with their everyday life. By large, the typical Electrolux customer values products that offer a premium, user-friendly and sustainable experience. Also, the Electrolux consumer is naturally curious and seeks new experiences. In regards to demographics, Electrolux is neutral in terms of targeting men or women, however, they are slightly geared toward customers from the growing middle-class socio-economic background.

Benefits

The benefits of opting for Electrolux are many. Great tasting food, e.g., through ovens that cook at precise temperatures, clothes that stay in shape thanks to washing machines that wash with care, and improved well-being by vacuums and air purifiers that provide a clean and fresh environment (Electrolux Group, n.d.).

Personality

Forward-thinking, curious, caring, responsible, and inclusive. These are some of the words that can capture Electrolux's personality as a brand. Our perception is that Electrolux drives and inspires change for the better, helping to improve the quality of people's life and the planet.

Values

These following drivers form the basis of Electrolux's whole operation: "Act sustainably, Create better experiences, and Always improve" (ibid.). They serve as the foundation for Electrolux's vision and mission to rest on. Electrolux places emphasis on sustainability and is determined to shape living for the better of its customers and for the health of the planet. They seek to deliver outstanding consumer experiences that enrich everyday life and interactions with products and services. They question the old ways of acting and are constantly searching for new ways of improving and see opportunities where others see hinders.

Mission

Electrolux states the following mission: "Reinvent lifetime taste, care and wellbeing experiences for more enjoyable and sustainable living around the world" (Comparably, n.d.). Taste, care and well-being are the three "innovation areas" which the company's offerings are divided into. "Reinvent lifetime" is linked to continuously improving and shows that the company is not set in their ways forever. Also, this implies that the lifetime of products needs to prolong and that our society and planet are constantly changing, and so needs Electrolux as well.

Vision

Electrolux states the following vision: "To shape better for the living, and to become the world's leading manufacturer of domestic appliances, as referenced by all stakeholders and the planet" (ibid.). This vision is a testament to the company's large ambitions, especially the phrasing "by all stakeholders and the planet", which shows the complexity of the challenge since these may have opposing interests and needs.

4.3 Competitor analysis

The results from the Competitor analyses are presented in this subchapter. Notably, both Bundles and BlueMovement currently operate solely in The Netherlands.

Bundles

Bundles is a service based in the Netherlands where new or refurbished coffee machines, dryers, washing machines and dishwashers can be leased for a monthly fee by a subscription (Bundles, n.d.). Bundles partners with Miele for the major domestic appliances and refurbishes and upgrades the machines before handing them off to the next user. This is assumed to facilitate the development of a more circular design (Holland Circular Hotspot, n.d.).

The pricing for the WMs consists of an initial deposit fee and a variable monthly fee, depending on which model is chosen and whether it is a new or a refurbished machine. The user is free to

cancel the subscription at any point, although if it is cancelled within 5 years, the user must pay a hefty return fee for the machine and will not recoup their deposit.

The service includes free delivery, installation, maintenance, and repairs, although the customer must also pay per use in addition to the monthly fee. The price per use varies, depending on how many times per month they, for example, wash. The more the user washes, the less they pay per wash. This could potentially lead to a negative impact on the sustainability potential of the business model as it may encourage the user to use the machine more than required, hence consuming more water and energy. To measure the use cycles online, Bundles connect a plug to the power outlet where the machine is installed as well as a bridge to the user's router upon installation.

The company offers two different Miele washing machines, which mainly differ by the higher tier model having a higher weight capacity, revolutions per minute, more programs and auto-dosing function. Despite the fact that the higher tier model has an auto-dose function, the service does not include the washing detergent in the pay-per-use model, which means that the provider could have a higher incentive to innovate for a lower detergent consumption if the PSS business model was of the functional result category.

BlueMovement

BlueMovement is a subscription-based service available in The Netherlands where customers lease appliances by Bosch or Siemens at a fixed monthly rate. The product offering includes washers, fridges, vacuums and many more. The customer may choose between either a new or refurbished product upon order. Furthermore, the leasing package includes servicing, repairs and replacement. Heavier appliances, including Washers, have a minimum subscription of one to six years, and depending on the duration of the lease, the lower the monthly rate becomes (BlueMovement, n.d.). Moreover, the service currently offers three washers, namely the Series 4, 6 and 8. The Series 8 comes equipped with Bosch's self-dosing system, "i-DOS". Deposits are not needed, albeit BlueMovement charges a one-time fee for set-up and installation, which is refundable within 24 hours.

Overall, BlueMovement does not differ substantially from Levande. For example, both services include refurbishment, installation, and like-for-like replacements. The subscription duration is the same, albeit WMs have a minimum of one year. Likewise, both services are categorised as Use-oriented PSS based on the Product-lease model. No additional pay-per-use models or similar are used currently.

4.4 Interviews

The chapter contains the results of the interviews. First, the interviews with Electrolux are described, and second, the ones with consumers.

Interviews with Electrolux

We held the interviews between February and March 2023 in a semi-structured format at an online video conference or at Electrolux HQ in Stockholm and were 35-60 minutes long. The list of respondents is presented in table 1.

Table 1. Interviewed Electrolux employees

Job position	Setting	Interview type
Project Manager for Levande	Online	In-depth
Senior Category Manager for PaaS	Electrolux HQ	In-depth
Global Design Care Director	Online	In-depth
Service Design Lead	Electrolux HQ	In-depth
Care Experience Development Specialist	Online	In-depth
Master Feel, Fit and Finish Designer	Electrolux HQ	In-depth
Feel, Fit and Finish Designer	Electrolux HQ	In-depth
Head of Circular Economy and Partnerships	Online	In-depth
Senior Industrial Designer	Electrolux HQ	In-depth
Design Director, Head of Research	Electrolux HQ	In-depth
	Total	**10**

Regarding the refurbishment of washing machines, from the interview with the Project Manager for Levande, it was discovered that the entire drum of the washing machine has to be removed for cleaning during the product's reconditioning for the next user, regardless of the level of use. This was seen as a major hurdle in the process and certainly the most time-consuming step of the procedure. Furthermore, several interviewees conjectured that increased wear and tear could be expected due to the more extensive use. Hence, it was suggested that a bespoke machine for PaaS should probably be more durable than conventionally sold machines. In addition, allowing servicing at people's homes would bring savings in terms of logistics. However, it was noted that keeping this process tidy and efficient is important since it may hamper the consumer experience otherwise.

During several interviews, the importance of maintaining the machine properly during the use cycle was brought up as machine breakdowns, odours and mould stains on clothes often can be linked to poor maintenance. Some of the most commonly broken WM components associated with neglected maintenance include the heater, the drain pump, the rubber bellow and the drum bearings. User maintenance consists of performing maintenance washes once a month, i.e., running an empty machine on the cotton program at the highest temperature, venting the drum after each wash, cleaning the drain pump filter, cleaning and venting the detergent drawer and its compartment, cleaning the rubber bellow, and lastly, performing a deep cleaning with "Electrolux clean and care powder" around four to six times a year, depending on usage. The information on how to perform maintenance can be found in the user manual. From the interview with the Care Experience Development Specialist, it was also found that using too much detergent leads to residuals being left in the machine's system after the program has finished, which is a good environment for bacterial and mould growth, causing bad odour and may leave stains on white garments.

Additionally, the interviewees brought up the benefit of PaaS in nurturing a deeper and more sustained relationship with Electrolux's customers and creating foundations for new approaches to enhance the customer experience. Suggestions in this area included connected services that inform the user about maintenance, monitor the use and continually improve the product through over-the-air software updates. Also, the possibility of performing prognostics to predict component failure was seen as a future area which should be investigated further. Proactively replacing parts and determining when they are likely to fail would help avoid more prominent issues in the future and facilitate on-site repairs. Several years back, Electrolux initiated a fixed repair price for their sold major domestic products, so ensuring that the repairs are cost-efficient would also benefit non-PaaS products. The company rates the importance of ensuring that the consumer feels taken care of highly, pointing to the benefit of similar PSS models, which alleviate the user's uncertainty of ownership, e.g. high initial fee and sudden servicing costs.

The service Levande had only been piloting for around six months. Trends that could be noted thus far were that consumers look for high-end machines with better capabilities with many functions and features. Although, if this was applicable in the context of Sweden should be investigated further.

Electrolux is devoted to user experience research, and it is common for the team to outsource researchers to fulfil their scope. However, interviewees mentioned that outsourcing is costly and often delays the development process, which is time-consuming and usually takes several years. Utilising a PaaS business model by direct customer interaction instead of interacting through third-party distributors was seen as an advantage. It could lower the need for outsourcing research as more insights can be collected directly from the consumer.

Moreover, the interviewees signified the strength in offering high-end features and performance since a unique selling proposition of PaaS is that it allows consumers to experience goods and services which may be too expensive to purchase. Overall, the respondents agreed that all actions which mitigate the risk of premature obsolescence are welcomed since, in the context of

Product-lease PSS, the washing machines are still owned by Electrolux. Therefore, enabling the product to stay in the service for as long as possible is valuable from a financial and environmental standpoint.

The interviews showed that the design team mainly uses a non-linear Design Thinking process when initiating new projects. The main focus and scope are to offer the consumer an outstanding experience; hence the user-facing areas are given the most attention. The other non-user-facing areas, such as the internal components, structural integrity, and the product assembly process, are free to be decided by the R&D team as long as they fulfil the design team's requirements and do not interfere with the user-facing areas. Consequently, the interviews showed that issues had occurred because secondary users, such as technicians, struggled to disassemble certain parts since they were not considered during the design process. In some cases, certain machine parts had to undergo a costly redesign process to allow an effective disassembly.

Interviews with primary users

We conducted the interviews from March to April 2023, and the list of interviewees can be seen in table 2. The interviews were held in person or via an online video call and were 30 to 60 minutes long.

Table 2. Interviewed primary users

Occupation	Age	Gender	Housing type	Tenure status	Setting	Interview type
Physician	29	Woman	Apartment	Tenant	In person	In-depth
Communications manager	56	Woman	Apartment	Tenant-ownership	In person	In-depth
Sommelier student	23	Man	Apartment	Tenant	Online	In-depth
Shop assistant	27	Woman	Apartment	Tenant	In person	Contextual
Editor-in-chief	33	Man	Apartment	Tenant	In person	Contextual
Entrepreneur	64	Man	House	Ownership	In person	In-depth
Financial assistant	62	Woman	House	Ownership	In person	In-depth
Pensioner	88	Woman	Apartment	Tenant-ownership	In person	Contextual
Pensioner	73	Woman	Apartment	Tenant-ownership	In person	Contextual
					Total	**9**

Regarding maintenance of washing machines, several interviewees mentioned that they were largely unaware of what routines were necessary for washing machines, especially compared to other major domestic appliances. One of the interviewees confessed to not having any maintenance procedures in place at all for WMs, conjecturing that both the dishwasher and dryer manage to be more evident with their maintenance needs. Another interviewee mentioned that they only discovered that the drain filter needed cleaning after it had clogged and forced an error code on the machine, which later resulted in the component needing replacing. This issue could have been avoided if the interviewee had been aware of the importance of said routine.

The interviewees mainly cleaned areas of the machine where stains and smudges were immediately visible, especially the detergent drawer but also the rubber bellow, although the latter needed to be folded out to gain a full view, see figure 14.



Figure 14. The rubber bellow folded out to gain access to the area needed to be cleaned

Parts of the machines which are less conspicuous are less likely to be cleaned, e.g. the compartment for the detergent dispenser drawer, or the drain pump filter, see figure 15.



Figure 15. a) The compartment for the detergent dispenser b) The drain pump filter being cleaned

One of the interviewees performed maintenance by cleaning the drain pump filter, as also shown in figure 15. First, a lid covering the filter had to be opened, followed by turning the filter 180 degrees and pulling it out. An observation was that the filter was difficult to turn at first, compelling the use of much force and switching hands mid-way. Notable was the water mixed with dirt running out on the floor, and the interviewee was heard groaning and complaining over the sight. It was found out that the machine had a drain hose that could be used for draining the water ahead of opening the filter, although it was close to the floor, requiring a bowl or bucket with short edges.

Moreover, several interviewees stated that they often do not read the instruction manual, which is still the primary touchpoint for finding information on maintenance. According to one respondent, their willingness to read manuals depends on their curiosity about the product. The person mentioned that they enjoyed reading the instruction manual for their TV since he was curious and excited about its many functions. This supports the notion that washing machines are not seen as enticing and are often overlooked compared to other home products. This is echoed by a quote from another respondent, sharing her initial thoughts on her WM: "It is relatively stylish, not the prettiest but also not the ugliest, but it is a washing machine".

Most interviewees found it difficult to dose the detergent correctly or did not pay attention to how much was needed for different amounts of clothes and mainly dosed by feeling directly from the detergent package. Most washing machines were installed alone or with a dryer stacked on top or next to it, see figure 16.



Figure 16. a) Dryer stacked on top of a WM b) Dryer standing next to WM

Several interviewees also proclaimed that sound alerts from washing machines are intrusive and a source of frustration. The interviewees questioned the reason why users must be alerted that the washing cycle has ended. This suggests that consumers prefer information to be given in a polite manner, which ties into how users should be informed about maintenance routines. Numerous interviewees were partial towards being instructed encouragingly, citing positive messages as a catalyst for performing maintenance.

Furthermore, several respondents were uncertain of the differences between washing programs and mainly opted for the same program, albeit altered the duration on occasion. The respondents conjectured that they would like to be provided with more information on the benefits of choosing specific programs and differences in terms of water and electricity consumption. The interviews showed mixed opinions on which factors contribute to energy and water consumption. Some interviewees liked to decrease the duration of the wash cycle in the hope of reducing total energy consumption while neglecting the issue that the machines need to be heated to temperature more rapidly, thus increasing the use of electricity. When asked about acquiring a used domestic appliance, the interviewees were sceptical since it came with uncertainties in terms of functionality, absence of warranty, fear of repair costs and freshness of the product. However, most were positive towards second-hand appliances and agreed that they perceived it as a more sustainable option, that a known brand was essential to gain trust and that a previously used machine that has endured several users indicated high quality.

Regarding leasing expensive products or when a major appliance is included in the tenantship, most interviewees said they value them as their own and would treat them more carefully. The reason for that was mainly a concern for high hidden service costs and penalty charges, or as one interviewee put it: “I would probably be more careful with it. I would not want to be the first one to scratch it".

Lastly, the interviewees that previously had leased products appreciated when the service was convenient and hassle-free and that it gave them access to products that they usually could not afford. It was expected that the leased product should function reliably and that it should include possible installations and quick repairs.

4.5 Customer journey map

We constructed the journey map visually in Miro and it can be seen in Appendix D. Table 3 shows the most prominent pains, gains and opportunities related to the consumer journey.

Table 3. Summarised CJM

	Before wash	**During wash**	**After wash**
Pains	● Bad smell from the drum ● Dirt in visible areas ● Correct dosing ● Worry about the seriousness of the machine's issue	● Loud noise and vibration from the WM ● Estimated duration of the program is incorrect	● Stained clothes from detergent residue ● Shrinked and/or damaged fabrics ● The machine has moved due to vibrations
Gains	● Appreciation for the variety of programs and adjustable settings ● Relief that the WM's issue is not serious and can be resolved	● Notices that the program initiates ● Now able to perform other tasks	● Fresh smell from laundry ● Removed stains
Opportunities	● Redesigned dispenser for easier dosing and cleaning ● Auto dosing ● Maintenance reminders ● Information about energy consumption ● New material or coating on the drum's exterior	● Quieter machine ● WM can weigh laundry clothes for more correct duration	● Wheels on machine for manoeuvring ● Ventilated WM door ● The door pops open automatically

4.6 Product autopsy

We conducted the autopsy with a Specialist Service Technician at Electrolux HQ in Stockholm. The session was held in the repair centre at Electrolux HQ in Stockholm and consisted of a detailed walkthrough of servicing of previous and existing machines. The autopsy highlighted several areas for improvement of the WMs regarding repairability.

First, an issue with the existing machines on offer is removing the front panel, see figure 17. In its current state, the disassembly process requires both the top panel and interface to be detached to reach the first screws holding the panel. After that, the machine must be tilted to access the last screws holding the panel together. This is troublesome ergonomically for the technician, who is required to lay down to access the fasteners, which also poses a danger of the machine tipping over the person. In addition, dryers are often stacked on top of WMs in Sweden and need to be lifted off first to perform repairs. Replacement of the pump is typically done via the rear of the machine and requires the technician to reach it while lying on the floor, despite the component being located in the front. The reason is that the front panel is deemed too tedious to remove.

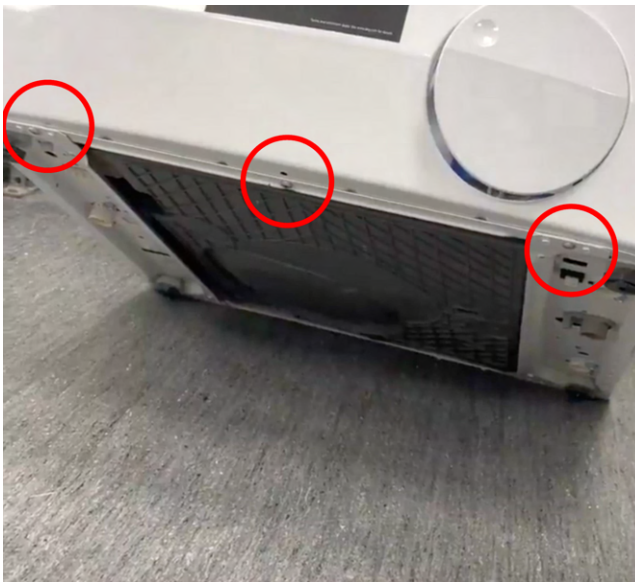


Figure 17. The bottom screws for fastening the front panel (shown in red)

According to the technician, issues with the drain pump are responsible for approximately half of all service calls, often due to insufficient cleaning or blockage by objects left in the pockets before washing, see figure 18.



Figure 18. Dirt was removed from the drain pump filter

Second, an issue was found on the expensive models concerning the front door, namely the hidden screws behind the inside door panel, see figure 19. Although not highly common, the door latch sometimes needs replacing, and the plastic panel must be removed to access the screws underneath. The technicians lack a bespoke tool for removing the panel and rely on a sharp object to pry it off. This is deemed as challenging, according to the respondent, and poses the risk of damaging the plastic. Note that this is also a part of the machine visible to the user, who may notice cosmetic damage after servicing. The supposed reason for hiding the fasteners in this manner is to achieve a cleaner design, however, it impedes the repairability.



Figure 19. The inner plastic door panel covers the screws and door latch underneath

The trend of removing screws is evident in the tub which encapsulates the drum also. For instance, the tub used to be separable in two halves so the drum could be cleaned easily. More than thirty screws fastened the tub's two halves. The supposed reason for changing to adhesive fastening is to save time on assembly and disassembly, see figure 20. However, the tub and the drum are now permanently joined together, i.e. the subassembly has to be replaced as one unit.

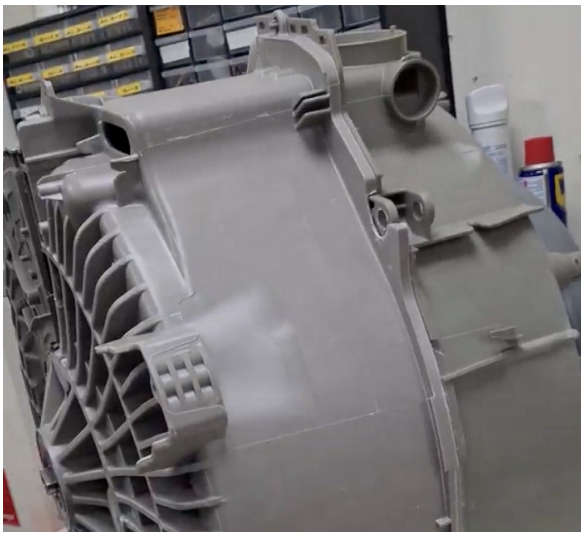


Figure 20. The two halves of the plastic tub are permanently joined

Third, according to the technician, a standard procedure on service calls is to run a machine diagnostics protocol at the start of each visit. This requires a laptop computer and a set of data cables to be brought along for plugging into the machine's control unit. Annoyingly, the top panel has to be removed for plugging in the cables since no port is found on the machine's exterior, see figure 21. Immediate access to the control unit would save unnecessary labour time on accessing the component. Moreover, the cables for running diagnostics are not standardised across the product range. The technicians must bring a large set of cables since it is difficult to know beforehand which one is compatible. Also, the technicians need to bring a laptop to the consumer to access the machine diagnostics program. The printed circuit board (PCB) is often replaced in this stage too, which is either updated from delivery or requires it to be done manually by the technician.

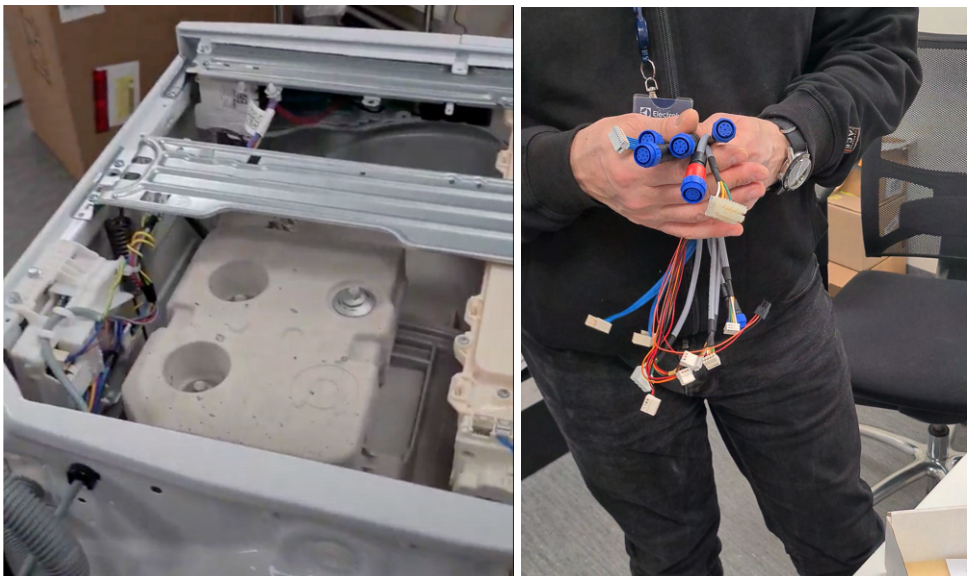


Figure 21. a) The top panel has to be removed to access the machine's control unit b) The cables for performing diagnostics are not standardised

The next issue pertains to the tools and fasteners used for disassembling the machines. Today's technicians require a set of three different sockets for untightening the screws for no apparent reason. This ought to be easily fixed, saving time and effort in dismantling machines to access components.

Furthermore, the most challenging service procedure is to remove the drum since it requires most of the other components to be removed before, and it was very heavy due to the attached concrete blocks used for stabilising the machine, see figure 21. Although not very common, the technicians replace the drum occasionally; for instance, due to complaints of bad smell. Lastly, according to the technician, installations and repairs are intertwined since improper installations may incur issues with components breaking prematurely, e.g., if a washing machine is not levelled correctly, it can vibrate excessively.

4.7 Participant observation

The observation consisted of four installations of washing machines in residencies in Stockholm, all conducted by the same installer. Both dialogue and aid from the observer were kept low to mitigate the risk of influencing their practices, albeit the installer was sometimes asked to clarify parts of the work and their experience performing it. Note that the person observed was an employee of an independent appliance installation company employee, contracted by several retailers and not affiliated directly with Electrolux.

First, the machines are unloaded from the vehicle for delivery to the consumer's home. The machines are then loaded onto a trolley and secured with ratchet straps, see figure 22. The installer carefully secures it to prevent it from wobbling during transport. The machine can then be lowered to the ground easily, thanks to the vehicle's hydraulic ramp.



Figure 22. The machines are loaded onto a trolley

Second, the machine is transported to the apartment by elevator if available, see figure 23. Note that the installer placed a hand in front to avoid the machine tilting forward. The clearance is measured, and on this occasion, the WM could not fit through the opening. Hence, the machine had to be carried via the stairs, which added tedious work for the installer and time spent on transport.



Figure 23. a) The WM is carried to the elevator b) The entrance to the elevator is measured c) The WM may have to be carried up and down the stairs

Third, the WM is unpackaged outside of the consumer's home door, figure 24. The main reason is to avoid litter inside the home. The transport plugs which secure the cables during transport also have to be removed. However, the bespoke tool was missing, which required the technician to use a plier, incurring extra time in this step by using a generic alternative. After removing the plugs, the next step was to remove the styrofoam piece at the bottom of the machine. The

installer had to lean the machine forwards, and the styrofoam persisted in moving with the machine, rendering it difficult to slide the trolley underneath.



Figure 24. a) The transport plugs are unscrewed b) And removed with a plier c) The WM is loaded back onto the trolley

Next, the old machine is removed from the home, see figure 25. The cables need to be fixated, however, the machine lacks anchor points since the transport plugs are no longer fitted. To solve this, the installer tied the power cord through the back of the detergent compartment. Additionally, it is common to find dryers stacked on top of the washing machine in Sweden, and they need to be removed to make way for the WMs. The door openings are also narrow and, therefore, easy to collide with objects and walls.



Figure 25. a) The cords and hoses are difficult to secure before transport b) The space inside apartments is often cramped

Subsequently, the machine is ready to be installed, albeit sometimes the installer has to check if there is a potential blockage in the water piping first. Foremost, washing machines are heavy and unwieldy to move around, see figure 26. The installer is forced to push the machine with their knee, which risks damage to the front. Once in position, it is also difficult to see behind the machines, which causes the person to search with their hand for the necessary touchpoints, e.g. the water hose and power cord.



Figure 26. Washing machines are unwieldy to move and difficult to see behind once positioned

In addition, when finding themselves at the back of the machine, a common issue pertains to the flanges on the screw for the inlet hose. Occasionally, there is insufficient clearance between the water pipe and the wall for the screw to be fastened, see figure 27. To solve this, the installer is forced to cut the flanges with, e.g. wire cutters. Moreover, in this sequence, the installer also fastens the lids, which cover the holes left by the transport plugs. These require a large force to snap in place, which could be made easier.



Figure 27. a) The flanges on the inlet hose screw need to be cut to be fastened b) The lids which cover the holes for the transport plugs require a large effort to be fastened

Once the machine is connected, arguably the most challenging stage is ensuring the machine is level with the ground. An unbalanced machine may lead to excessive vibrations and damage to components long-term. The installer uses a spirit level to measure angular deviation in two directions and adjusts the position by screwing each of the four feet individually, see figure 28. An issue is that the machine must be tilted back to access the feet. Electrolux used to deliver the machines with a bespoke wrench which was thin enough to slide underneath the machines without tilting them backwards, but unfortunately, they are no longer offered. The rear feet are even more challenging to reach, especially if the machine is placed below a bench. The challenge is exacerbated further if the machine is placed near a floor drain since the ground also slopes.



Figure 28. Ensuring that the machine is level is vital when installing

Lastly, before leaving the consumer, the installer verifies that it functions correctly by running a quick program. The installer will also take the time to demonstrate the machine to the consumer if they wish.

4.8 Analysis on the wall

We categorised the insights according to WM routines, WM functions, Used products, Product leasing, Consumer interactions, Maintenance, Installations, Repairs, and PaaS for Electrolux. Table 4 presents the summarised insights listed under each category. Note that the insights are based on information from all research, not solely from interviews with e.g. consumers.

Table 4. Summarised insights from analysis on the wall

Category	Summarised insights	Source
Laundry routines	Users do not know how much detergent is needed and mainly dose on feeling	Interviews with primary users
	Consumers seldom read the printed manual and therefore miss information on maintenance and features	Interviews with primary users
	Consumers do not understand the difference between different WM programs and how much water and energy they consume	Interviews with primary users
Used products	The perception of the brand is important when choosing a second-hand machine and establishes trust for the product	Interviews with primary users
	A previously used machine that has endured multiple user cycles is seen a token of quality	Interviews with primary users
	Consumers have a positive attitude towards using second-hand domestic appliances and consider them more sustainable, granted that they are fresh and they can trust that they perform as well as a new product	Interviews with primary users
Product leasing	When leasing expensive products or renting a residence where major domestic products are included, users value the products as their own and treat them more carefully	Interviews with primary users
	Consumers are frustrated by hidden costs and fear expensive penalty charges if a leased product would get damaged	Interviews with primary users and Electrolux, (Lieder et al., 2018)
	Users like to lease products because it is convenient, hassle-free, and enables access to products they normally cannot afford	Interviews with primary users, Electrolux, (Lieder et al., 2018)

	Consumers expect the machines to be functioning reliably	Interviews with primary users, Electrolux, (Lieder et al., 2018)
	Consumers expect the service to include installation and quick repairs	Interviews with primary users, Electrolux, (Lieder et al., 2018)
Maintenance	Most consumers neither know that maintenance is needed nor how to perform it	Interviews with primary users
	Consumers mainly clean parts where dirt is directly visible, such as the detergent drawer and the rubber bellow	Interviews with primary users
	Parts that need maintenance need to be easy accessible and easy to clean	(Van den Berg and Bakker, 2015)
	Consumers do not appreciate maintenance alerts that interrupt the user in their everyday routines	Interviews with primary users
	Consumers do not want to be forced to do maintenance	Interviews with primary users
	Consumers have a positive attitude towards receiving informative, friendly and encouraging reminders of maintenance	Interviews with primary users
	Poor maintenance can lead to odours and premature component failure	Product autopsy, Interviews with Electrolux
	Consumers want to be made aware of and prepared for maintenance routines	Interviews with primary

		users, (Ceschin and Gaziulosoy, 2020)
	Endowing products with personality can strengthen the emotional bond to them and foster caring behaviour	(Ceschin and Gaziulosoy, 2020)
Installations	WMs are unwieldy to move and may cause damage to the consumer's home during transport	Participant observation
	Levelling of WMs is challenging because of tight spaces and difficult access to the machines' feet	Participant observation
	Improper installation may lead to premature failure of components	Product autopsy
	Loose cords and hoses are difficult to secure and may snag onto the surroundings during transport	Product autopsy
	Consumers want to trust that the installation has been done correctly before the installer leaves the home	Participant observation, Interviews with primary users
Repairs	Washing machine development projects do not consider secondary users such as technicians, causing expensive post-production modifications and time-consuming repairs	Interviews with Electrolux
	Hidden fasteners for technicians lead to time-consuming disassembly	Product autopsy
	Access routes to internal components are inefficient on Electrolux' current machines	Product autopsy
	Diagnostics currently need to be performed on-site and are difficult to perform due to limited access and need for unstandardised connectors	Product autopsy
	The amount of fasteners should be minimised for efficient disassembly	(Van den Berg and Bakker, 2015)

PaaS for Electrolux	Durable products with a long lifetime is essential for a successful business model	(Tukker, 2004)
	Connected machines can monitor the usage and are imperative for user behaviour and lifetime prognostics and preventing breakdown	Interviews with Electrolux, Competitor analysis
	Collecting consumer insights during the use phase fosters a stronger consumer relationship and is beneficial for the continuous improvement of the WM and the business model	Interviews with Electrolux
	Including and considering the installation and on-site servicing open up new business and design opportunities	Interviews with Electrolux
	It is advantageous for the product to be designed so that it can be used for other business models with a higher sustainability potential	(Tukker, 2004)
	The design department mainly focuses on user-facing areas and are usually involved in the early phases of the product development	Interviews with Electrolux
	The product should be designed to facilitate the use of the least energy consuming functions and programs	(Ceschin and Gaziulosoy, 2020)

5 PAAS DESIGN PRINCIPLES RESULT

This chapter presents the results for the PaaS Design Principle phase that we derived from the research, comprising five design principles encompassing the necessary considerations for PaaS. Furthermore, we evaluated the principles through a creative workshop.

5.1 First loop of brainstorming

The first loop of brainstorming generated ideas based on the summarised insights shown in table 4. Figure 29 shows a collection of the ideas generated by the brainstorming session. This ideation showed that certain summarised insights were difficult to relate to product design. For instance, the insights pertaining to product leasing were largely relevant for departments other than design teams, such as management, marketing or sales. Additionally, while certain insights from PaaS for Electrolux provided knowledge about the design processes, their link to product and service design was less apparent; the design department mainly focuses on user-facing areas and is usually involved in the early phases of product development. This result alluded to the fact that the insights needed to be developed further to make them more comprehensible, relevant and actionable.

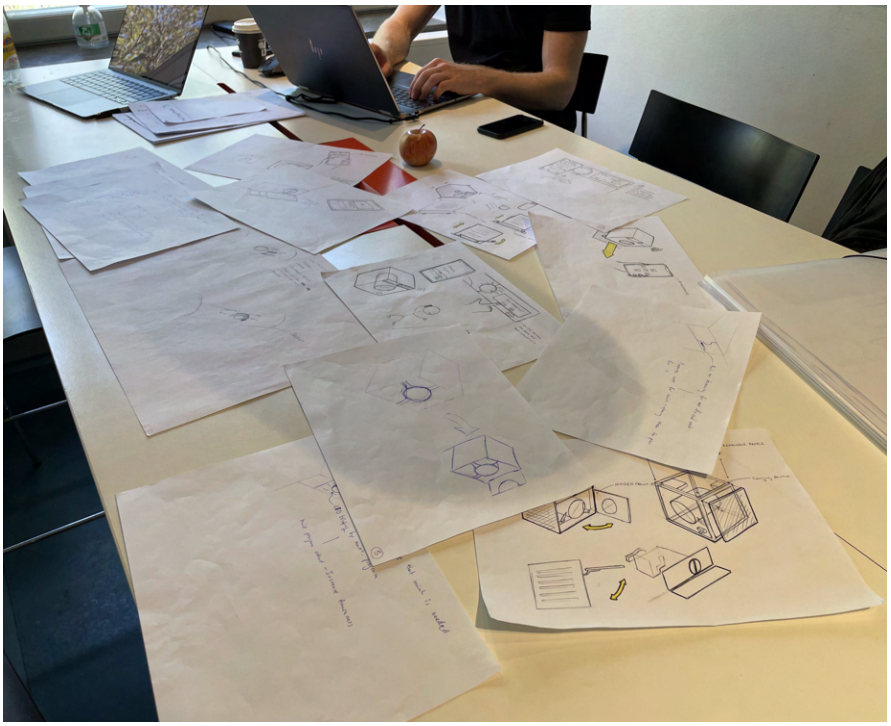


Figure 29. Collection of ideas from the Brainstorming session

5.2 PaaS Design Principles

Based on the first loop of brainstorming, we developed key insights and refined them into PaaS Design Principles. We combined the categories shown in table 4 based on similarities of their respective insights. This led to five new principles: Ensure maintenance, Trust and transparency, On-site repairs, Seamless interactions, and Convenient installations. Figure 30 presents an overview of them.

The PaaS Design Principles are an addition to Electrolux's current design prerequisites and complement how designers should think when designing for PaaS. The principles were purposefully phrased to exhort the designers to be creative and not restrain the designer with strict demands. Each principle provides a brief background and context for the designer with the purpose of building a holistic understanding of the necessary considerations for designing washing machines for PaaS. The considerations are formulated as actionable design opportunities which can be used in ideation sessions for the design team.

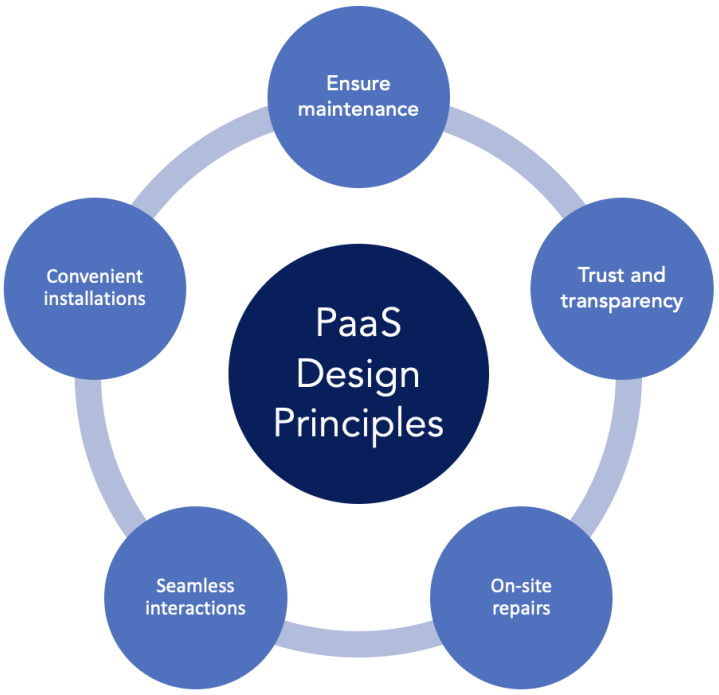


Figure 30. Overview of the PaaS design principles

5.2.1 Ensure maintenance

To prolong the machine's life and prevent breakdowns, it is essential to maintain the WM regularly. This includes doing maintenance washes, descaling, and cleaning the rubber bellow, detergent drawer and filter. The research showed that most users rarely read the manuals and are unaware of WM maintenance. They only perform it if they notice something looks dirty or the machine can not be used due to an error. Strengthening the emotional bond to products to nurture

caring consumer behaviour and prevent functional and emotional obsolescence. The consumer wants a functioning, high-performing machine, and it is in the service provider's interest to prolong the life of washing machines in a PaaS system. The design should therefore make the user aware of maintenance by communicating:

- That maintenance is needed
- Which maintenance is needed
- How to perform maintenance

The users do not want to be forced to do maintenance as it will harm the overall experience with the business model and the leased product. Instead, they prefer to be encouraged to perform maintenance, and the solutions should therefore:

- Make the parts that need maintenance easy to clean and reach
- Make the user aware in a friendly and encouraging way
- Award the user for performing maintenance
- Prepare the user for maintenance

5.2.2 Trust and transparency

The research showed that many users are unaware of the differences between washing programs and functions of the washing machine, and that transparent and clear communication is integral in establishing trust for the service provider. By choosing a PaaS machine, it is possible to reduce the environmental impact and generation of e-waste from cradle-to-grave products. Consumers prefer sustainable solutions, thus, explicitly expressing the benefit of opting for refurbished appliances is vital. Presenting the energy consumption of washing programs is also welcomed by consumers and increases trust in the service provider. The consumer should clearly understand the advantages of leasing instead of directly purchasing to accept a subscription agreement. Finally, enable the consumer to adopt sustainable behaviour and increase their willingness to do so.

The principle can be condensed as follows:

- Communicate the purpose and benefits of programs clearly to consumers
- Communicate environmental gains to the consumer
- Communicate that the refurbished WM is equally as capable as a new product

5.2.3 On-site repairs

As mentioned, prolonging the lifetime of products is essential in the context of Circular Economy, and also for the provider since ownership is no longer at the hands of the consumer in both Use- and Result oriented PSS. This principle emphasises the importance of repairability for extending the products' lifetime and ensuring that the WMs can endure multiple user cycles. In order to guarantee effective refurbishing, components need to be disassembled easily without them breaking.

On-site repairs allow the WMs to stay in the hands of the consumer and save costs on transportation. It is also important to consider the technicians as secondary users and ensure user-friendliness for their role. Effortless repairability results in technicians who are content with their work environment and thus better equipped for delivering an excellent consumer experience when engaging with the consumers.

The principle can be summarised into the following areas:

- Easy access to internal components when the machine is installed
- Clearance for disassembly tools
- Minimise the amount of fasteners needed to reach the internal components
- Visible fasteners for technicians when the machine is upright
- Minimise the use of different tools needed for disassembly
- No permanent fastening of components

5.2.4 Seamless interactions

The consumer wants a functioning WM that aids them in their everyday life. When it breaks, it interrupts their daily routines as their laundry piles up, forcing them to book a technician, stay at home at a specific time during the visit and wait for the repair to be finished. The technicians rarely have all the components for all machines brought to consumers. They often have to do diagnostics on-site first, order the relevant components and do a second visit. The more frequently a technician needs to visit the consumer to repair one issue, the wearier the consumer becomes, decreasing their likelihood of wanting to keep the same machine instead of opting for a new one.

Continuously learning consumers' usage patterns and receiving direct feedback gives a basis for continuous improvements. The design should enable the product to stay relevant over time through upgrades and updates to prevent technological obsolescence and prolong its lifetime.

To summarise, consider the following points:

- Predict failure of components
- Ensure easy diagnostics
- Enable swift software updates and upgrades
- Learn users’ usage patterns
- Facilitate quick support for users

5.2.5 Convenient installations

Many apartment buildings in Stockholm are old with cramped space for transporting and installing machines. The research shows that it is cumbersome to transport the washing machine both on the way to the consumer’s home and inside; surrounding objects and walls, et cetera, may be damaged when colliding with the machine. The cords stuck onto objects easily, and it is challenging to position the machine correctly in terms of levelling and overall placement in the

laundry room. Within the context of PSS, it can be expected that the machines will be transported more regularly. Furthermore, at the end of each subscription, the machine passes through the loop of refurbishment and possible repairs until it is ready for the next user cycle. This stresses the importance of time efficiency to deliver on promises such as 48-hour replacements, which consumers highly value.

- Ensure cords do not get stuck during delivery
- Ensure the machine is easy and safe to transport
- Enable the machine to be positioned easily
- Prevent the machine from damaging consumers' homes
- Prevent damages to the machine during transport
- Ensure that the user is confident that the installation is correct

5.3 Creative workshop

We organised a creative workshop with participants affiliated with Electrolux with competencies from different fields. The purpose was to evaluate the design principles with the participants by ideating based on them to ensure they were useful in generating innovative PaaS solutions. The background of each principle was first explained, and then the participants were briefed on the considerations for designing for PaaS for each one respectively. The participants were then asked to ideate for five minutes and place them on a wall to present to the rest of the group, see figure 31. They were also asked to vote on their favourite idea after each session. Table 5 lists the participants of the workshop.

Table 5. Workshop participants

Job position
Senior Industrial Designer
Master FFF Designer
Senior lead, Circularity by Design
Service Design Lead
Circularity engineer
Total: 5

By and large, the group could generate many ideas based on the considerations listed in the principles and appreciated having the background of each principle written next to the bullet list of considerations. However, some comments regarding the connection between trust and transparency with PaaS arose since they were not entirely evident. The principle is important in areas other than PaaS, but the research shows that trust is essential for committing to

subscription agreements. Also, consumers are largely unaware of what WM features and programs entail and wish for transparency concerning energy consumption. However, it was possible to tell that the participants struggled more to generate ideas for this principle than the others. Nonetheless, they could generate ideas around, e.g. “Eco-scores” and new program layouts corresponding to the ones presented in chapter 4.1.

When asked if they considered the five areas currently in their design scope, the participants hesitated a bit before answering. This suggests that the principles bring new attention to areas overlooked in Electrolux’s current design process, especially the significance of installers and technicians as user categories. One participant, formerly working in the R&D department, conjectured that they currently have, e.g. repairability in mind. However, it is less prioritised than other metrics, such as manufacturing optimisation or retail data. One member of the group echoed that sentiment by saying that they had made efforts to reintroduce fasteners after facing criticism for difficult disassembly. One participant added to the discussion by highlighting that the issue of speaking of repairability can negatively impact consumer experiences since the word can have a negative connotation, i.e. consumers may associate the word with failure when deciding on a new purchase. Additionally, weak repairability has a negative effect on the consumer experience in the sense that technicians are out on the field speaking to consumers. Furthermore, if they think the products are subpar in terms of service, that opinion might influence consumers to choose a different brand.

Moreover, the participants mentioned that the industrial design team works extensively with the user-interface elements, notably the graphics printed on the machine. Currently, Electrolux prints the program names in separate languages for different markets, which is a laborious process. The group was interested in exploring means of presenting information innovatively, e.g., rearranging the programs according to energy consumption. This could be combined with increasing haptic feedback, making the knob harder to turn by order of consumption. Also, the group were intrigued by visceral feedback from traffic lights to communicate maintenance status. Likewise, ideas of endowing the machine with a personality for increasing maintenance by the consumer were intriguing.

Overall, the participants were positive towards the principles and appreciated that they were “actionable”, i.e., the insights were defined as clear design opportunities. The participants responded that the principles provided a solid overview of the necessary considerations for PaaS and showed promise in being used as a tool for future ideation sessions.


PaaS Design Principle
PaaS Design Principle
PaaS Design Principle

6 IMPLEMENTATION IN DESIGN RESULT

The chapter describes the implementation of the derived principles in design through ideation and a developed PaaS design proposal.

6.1 Second loop of Brainstorming

The chapter presents the ideas from the second loop of brainstorming. We performed this loop to generate ideas for the design proposal explicitly based on each of the five PaaS Design Principles. The brainstorming sessions focused on one principle at a time to maximise attention to each respective area.

Ensure maintenance

The five most promising ideas, according to us and in terms of their feasibility from the principle Ensure maintenance are shown in figure 32.

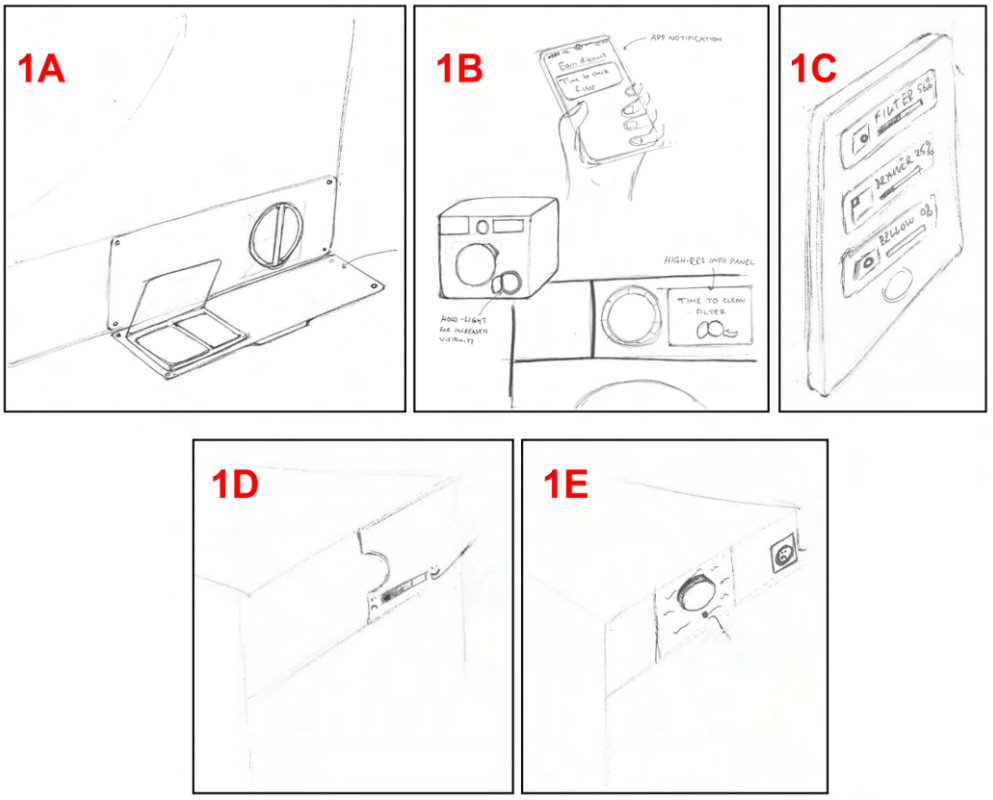


Figure 32. The main ideas from Ensure maintenance

1A is a service door where the detergent dispenser has been placed next to the drain filter to raise maintenance awareness by positioning the touchpoints of the machine close to each other. 1B is a colour screen on the machine which can show rendered graphics of maintenance areas for more explicit information. This idea also features a LED light around maintenance touchpoints, such

as the filter door, detergent dispenser and rubber bellow. Also, the incentive for performing maintenance is increased by offering discounted subscription fees after performing maintenance.

Next, 1C is an app with maintenance tutorials and notifications on maintenance and displays maintenance status on components which the consumer can reset manually. Finally, 1D and 1E are based on giving the machine a personality by showing either a sad or happy icon, depending on the machine's condition. The thought is that establishing an emotional bond with the appliance will advertently increase maintenance by the consumer.

Trust and transparency

The three most exciting ideas, according to us and in terms of their feasibility from the principle of Trust and transparency, are shown in figure 33.

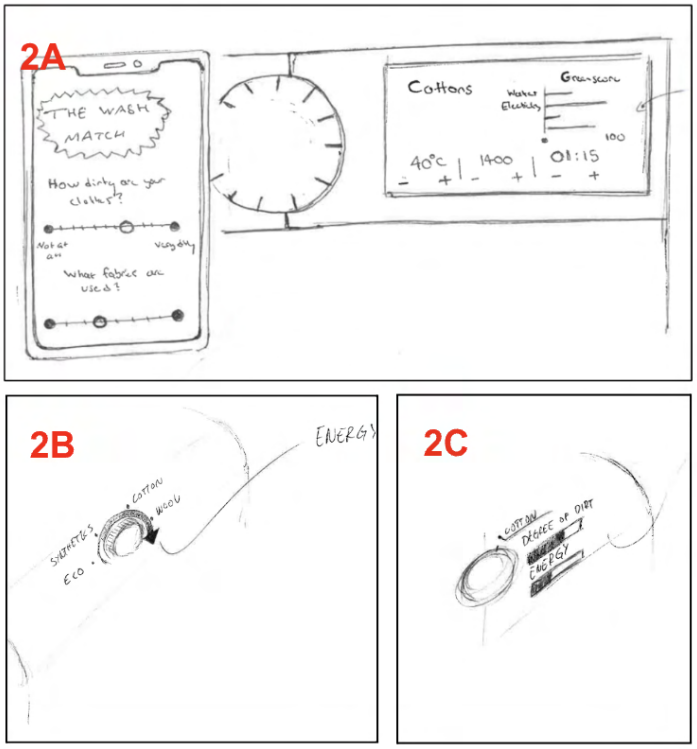


Figure 33. The three most promising ideas for Trust and transparency

2A shows a User Interface (UI) on the machine paired up with an app that shows a "green-score" rating for metrics, such as water and energy consumption, compared to the most consuming program. The idea is to make the program's functionality more transparent to ensure the consumer understands the differences between programs. The app also shows a "wash-match" where the consumer can fill in how dirty their clothes are and what fabric they are washing so that the system automatically can determine the best program for the user. 2B has a more straightforward graphic next to the knob, where the programs have been sorted from the least to the most energy-consuming. Finally, 2C shows an additional bar where the user can see the amount of dirt the program can remove from the clothes.

On-site repairs

The most promising ideas according to us and in terms of their feasibility based on the principle of On-site repairs, are shown in figure 34.

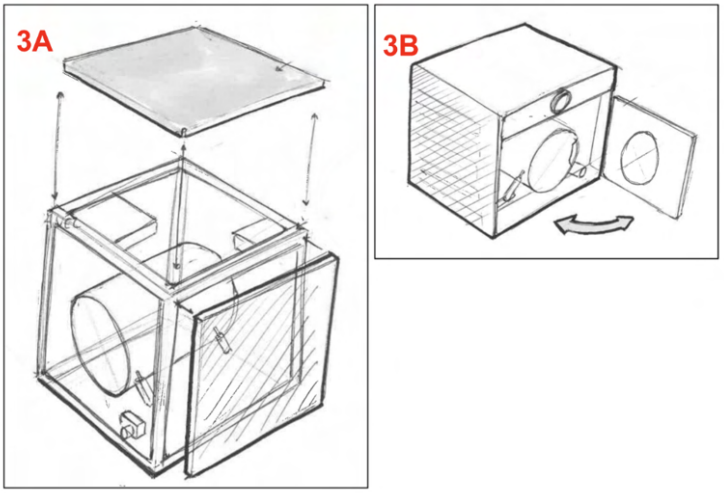


Figure 34. The most promising ideas for On-site repairs

Firstly, 3A is a cage structure with separate removable panels for each side. Hence, internal components are accessible from all directions. Secondly, 3B is a hinged front panel which enables repairs from the front.

Seamless interactions

The most promising ideas, according to us and in terms of their feasibility from the principle of Seamless interactions, are shown in figure 35.

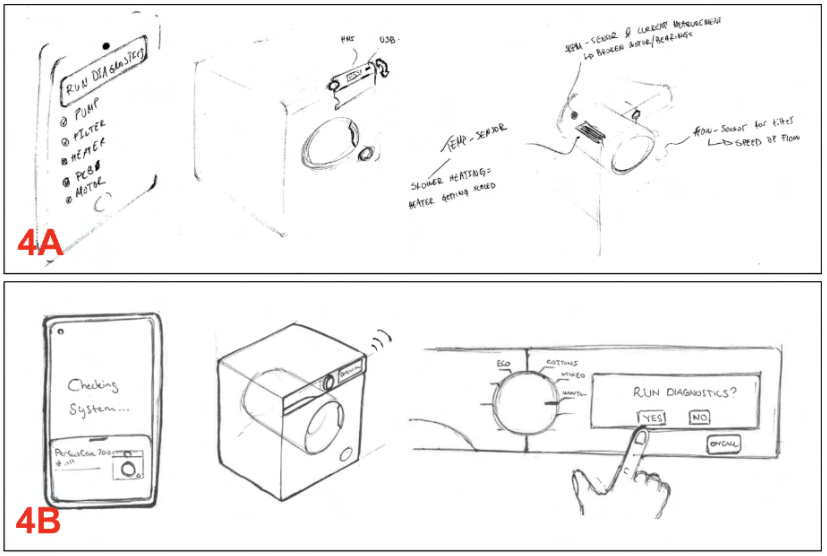


Figure 35. The main ideas from category PaaS for Electrolux

4A is a diagnostics app for technicians with sensors on critical components to diagnose their performance. The interface panel is also hinged to access the HMI board beneath easily. Here, the technician can find a USB port for plugging into the machine on the exterior, which mitigates today's issue of having to remove the top panel first.

Next, 4B is a diagnostics program on the machine which the consumer can initiate. If the machine detects a malfunction, help is automatically sent to Electrolux, which institutes a service call. This idea also features an "On-call" button, akin to the ones found on modern cars with connectivity, which requests help from the service provider when pressed.

Convenient installations

The ideas most promising ideas, according to us and in terms of their feasibility from the principle of Convenient installations, are shown in figure 36.

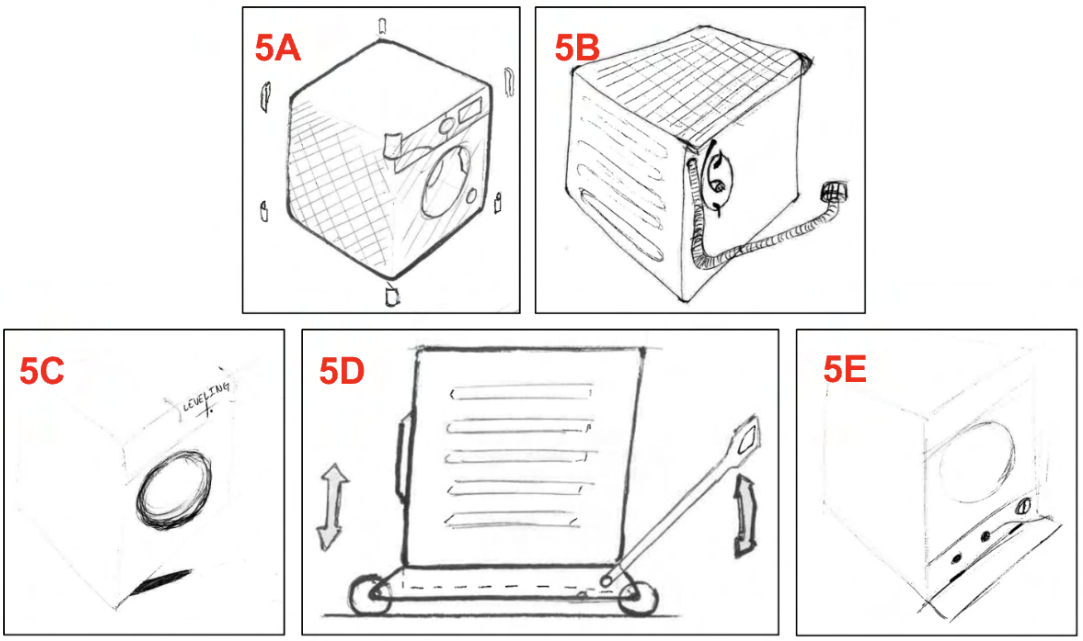


Figure 36. The most promising ideas from category Installations

First, 5A is a set of detachable rubber bumpers on the machine's corners to avoid damaging the consumer's home during delivery. Second, 5B is a cable reel on the machine's rear for securing the cords and speeding up uninstallation. Also, the screw on the water hose lacks the flanges found on the ones today, which ensures that they can be fastened without colliding with the wall.

Third, 5C is an installation program on the machine's display with an in-board level sensor showing the real-time position during levelling. Moreover, the program assures the consumer that the installation is done correctly. The penultimate idea, namely 5D, is a set of wheels fitted to the machine that increases the machine's moveability. The machine can be lowered once it is in its favoured position. Finally, 5E is a front adjustment of the rear feet, utilising the principle

found on dishwashers, i.e. a worm gear for transmitting power to the rear via a long screw spanning from front to rear.

6.2 Choosing ideas for design proposal

We selected ideas from each principle to develop a complete design proposal. The ideas were chosen by considering the summarised insights from the research to decide which idea fulfils each principle's considerations best.

Ensure maintenance

First, to ensure maintenance, the decision was made to fit a Colour screen to the WM's user interface, see idea 1B in figure 32. This offers more freedom when communicating with the consumer than today's pixelated display. The screen that can show computer graphics opens up a range of opportunities for informing the consumer of the touchpoints related to maintenance. For example, the screen can visually indicate the drain pump filter, ensuring the consumer understands its location and how to clean it. Moreover, moving the previously printed program names onto the display saves effort in translating them for different markets. Another benefit is that screens can be updated if new functions and programs are added. This aligns with EDD by ensuring the product stays relevant over time.

Furthermore, the decision was made to incorporate a maintenance status bar in the screen to prepare the user for maintenance, see idea 1D in figure 32. As mentioned, consumers neither want to be forced to perform maintenance nor should the WM cause an interruption in their daily activities. The final design change from this principle is adding a dedicated maintenance wash program. Namely, the wash cycle is designed for cleaning inner components such as the drum to refresh the machine and prolong its lifetime.

Trust and transparency

The decision was made to incorporate an Eco score bar to the display, see idea 2A in figure 33. Similar ideas showed promise by the participants in the workshop, and consumers wish for more transparency in terms of energy consumption. Moreover, Eco-score adheres to DfSB by enabling consumers to adopt sustainable behaviour through choosing eco-friendly washing programs. The research shows that consumers are unsure of what parameters affect their energy consumption. For example, many think that lowering the wash cycle duration saves energy. Instead, the machine has to heat the water more rapidly, which draws more electricity. The Eco-score visualises the impact of adjusting the program's duration on energy consumption. Additionally, in PaaS models, the environmental impact during the use cycle should be minimised since subscription models risk more intense use than usual. The other ideas shown in figure 33 are also interesting and can be explored further in future projects.

On-site repairs

Access to components from the front aids greatly with on-site repairability. A removable front panel mitigates the issue of de-installing the machine and pulling it forward to remove the rear

panel. Immediate access to, e.g., the drain pump saves considerable time and effort on lying on the floor and stretching all the way forward to reach it from the rear. Furthermore, cosmetic damages to the front are most noticeable, and a removable front panel can easily be replaced to give the machine a new appearance. Although promising from a design for remanufacturing perspective, the cage structure shown in figure 34 would require an extensive architectural redesign of the platform. Also, it would likely require more fasteners than the current design and more parts overall. This PaaS design proposal focuses on facilitating on-site repair from the machine's front whilst other access points are not prioritised.

Seamless interactions

To implement seamless interactions, the most significant changes are to add remote diagnostics, upgrades and updates of the WM. Per idea 4A in figure 35, the new sensors are fitted to the design proposal, thus enabling prognostics to be performed. The research shows that technicians sometimes do not bring their laptops during service calls, and accessing the machine's control unit requires partial disassembly of the WM.

Convenient installations

Since the most laborious step of the installations is levelling, the decision was made to incorporate the program idea 5C in combination with idea 5E (see figure 36), consisting of adjustable feet from the front to facilitate easy installations. The decision was made against fitting wheels to the machine according to 5D since easier levelling takes precedence over manoeuvring. Front-loaded machines are also subjected to intense vibrations during operation. Thus, the feasibility of wheels is uncertain when considering the risk of the WM moving while a program runs. It is doubtful whether breaks for the wheels could induce enough stopping power for the machine to avoid drifting away. Nonetheless, the idea is promising and may be investigated in the future.

6.3 PaaS design proposal

The PaaS design proposal was based on a high-end WM from Electrolux that featured a wifi-module and auto-dosing function. The research showed that high-end products are the preferred offering in PaaS business models. Foremost, Auto-dose ensures correct dosing each time. It also mitigates the issue of undissolved detergent and mould build-up in the machine's system due to excessive manual dosing of washing detergent. An added benefit of Auto-dose is that it enables pay-per-use models in the future, where the consumer is charged at a variable rate based on the usage level. Finally, a wifi module is necessary for reaping the benefits of connected WMs.

We developed the complete design proposal by combining the chosen ideas from all principles and adapting them to work together and to fit Electrolux's brand. The final design proposal was designed in Fusion360 (Autodesk, n.d.) and can be seen in its entirety in figure 37. The UI elements were designed in Figma (Figma, 2019).



Figure 37. Perspective view of the design proposal

Adjustable feet and installation program

The adjustable feet from the front were made possible by the principle found on many dishwashers, which feature a long screw spanning from front to rear and driving the vertical screws for the feet as worm gears, see figure 38.

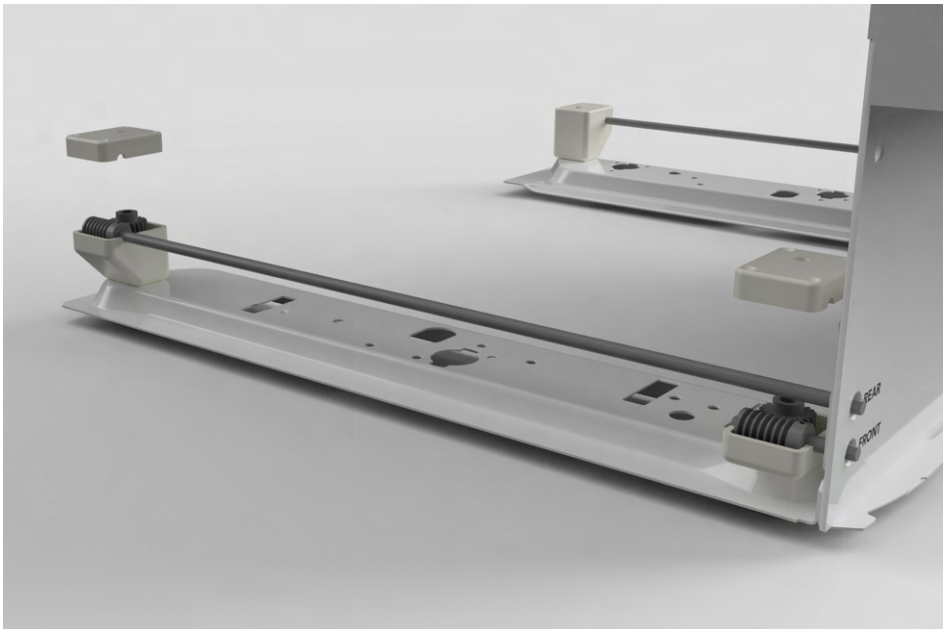


Figure 38. The feet are adjustable from the front by worm gears placed on each foot

To hide the screws from the user-facing front, it was decided to hide the screws behind a cover at the lower part of the front panel. This new cover also replaced the lid for the drain pump filter and was designed as a tray with raised edges. This means it can be placed on the floor and contain the water exiting the filter when draining it, see figure 39. The cover can be fastened back onto the front panel via tabs on either side and an additional magnet at the top.



Figure 39. a) The cover was designed to contain the water exiting the filter b) The cover snaps back onto the front panel

Furthermore, the PaaS design proposal consists of a dedicated installation program where the installer can follow a step-by-step program after connecting the machine. The machine is equipped with an accelerometer for determining its level, shown in the program's first step, so that the installer can easily adjust the machine when it is in place, see figure 40.



Figure 40. Installation program showing the machine’s level

The level position of the machine is determined by a dot relative to a crosshair. By adjusting the feet, the dot will move, e.g., if the machine is leaning forward, the dot will be above the horizontal line, and if it is leaning to the right when seen from the front, it will be to the left of the vertical line. It is used similarly to the classical labyrinth game, seen in figure 41. When the position is correct, the installer gets an encouraging confirmation message “You’re spot-on”.



Figure 41. The levelling is used similarly to the classical labyrinth game

The last step of the installation program is to check that the machine functions correctly. The program fills the drum and drains it, confirming that the machine is level and finally spins it. The idea is that the installer will run the program at the end of the installation to ensure that the machine is in order and that the installation has been done accordingly. This is pleasing for the installer and the consumer, who is assured that the machine operates as wished. Moreover, all four feet will be adjustable from the front without the need to adjust them underneath the machine.

EcoScore

The EcoScore consists of two bars showing water and electricity consumption on a colour touch display. The first idea was to include kWh consumption since the wash programs had a big difference in consumption. However, it was discovered that they had relatively low consumption overall and that the consumer would not be encouraged enough to choose the least consuming one. Therefore, the thought is that on an arbitrary scale, comparing the programs to the most consuming one is likely to nudge the consumer more effectively since the differences can be visualised more strikingly. In addition, the resulting EcoScore is given as a percentage, and the font colour from worst to best is red, yellow or green, depending on the rating, see figure 42. The crucial aspect is that this rating is updated directly when parameters such as temperature are altered.



Figure 42. The proposed UI design of the EcoScore

Maintenance bar

The maintenance bar features a bar to show the condition of the parts that need to be maintained. Warning messages are shown visually when the maintenance reaches low levels. This ensures that the consumer is given notice of upkeep early in time and not abruptly once the maintenance is required. The overall maintenance level is represented by an icon with a facial expression corresponding to the machine's condition. This is inspired by service bars on car instrument panels which show mileage left for the next service. The decision was taken to implement this functionality directly on the machines, the thought is that it will gain the most attention from the consumer. This works in compliance with EDD by showing a happy or discontent face on the machine, depending on the machine's condition, see figure 43.



Figure 43. The maintenance bar

The conceptual UI featuring the EcoScore and Maintenance bar is shown in figure 44. The UI features a clean graphical design with modern Sans-serif fonts, and the colours offer strong contrast against the black background. The blue accent colour was chosen to align with Electrolux's graphical identity.

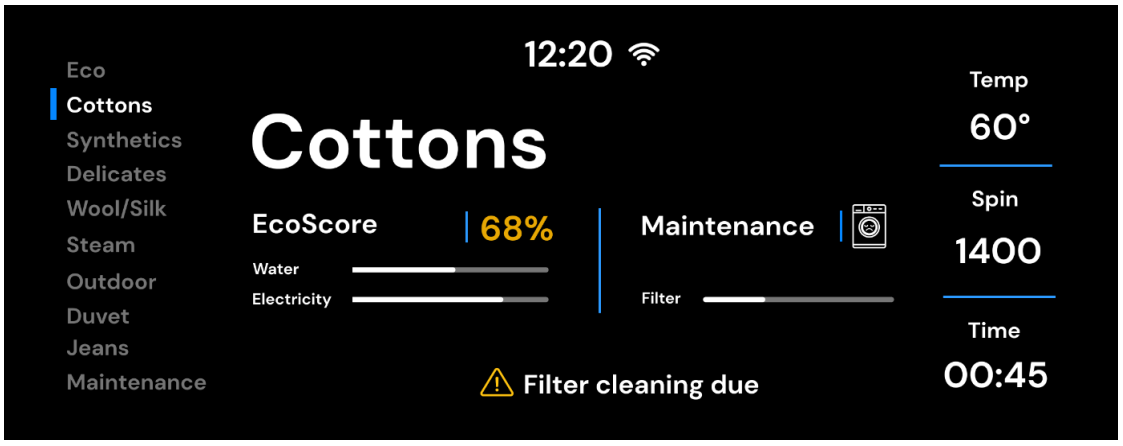


Figure 44. Conceptual UI on the machine's screen

Diagnostics and upgrades

To facilitate remote diagnostics and upgrades, we suggest incorporating a pressure sensor for the drain pump filter, which can detect malfunction proactively in the washing machine. The machines can be updated wirelessly thanks to their Wi-Fi connectivity, see figure 45.

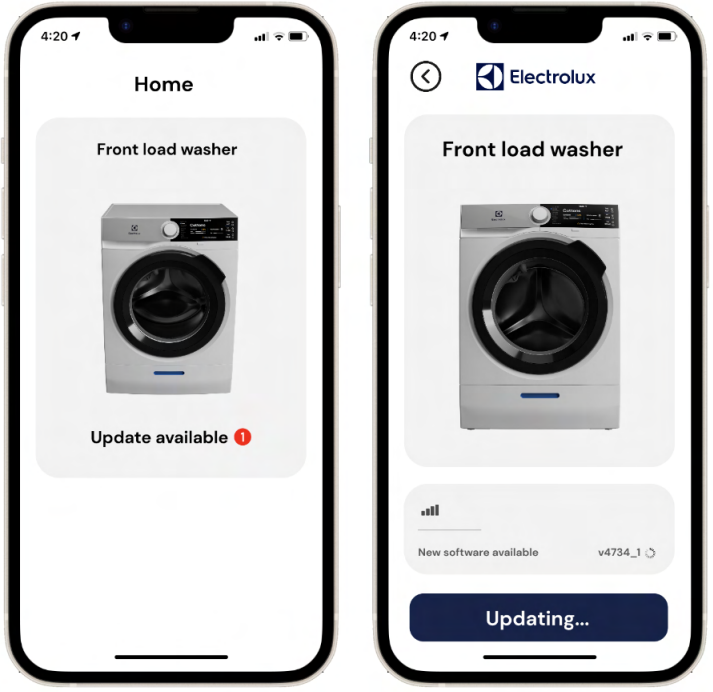


Figure 45. Over-the-air updates

Removable front panel

Firstly, to make the front panel removable, we suggest that three tabs are added to the bottom of the front panel, which correspond to three slots in the bottom bracket, see figure 46. This eliminates the step of having to tilt the machine back to access the screws from underneath. The lower two screw holes have also been brought down to make them accessible from the front. They are exposed after removing the cover as mentioned earlier.

Furthermore, the front panel is separated from the rest of the panels, i.e. the technician no longer needs to remove any other panels to remove the front. Thus, allowing the machine to stay in place and gaining the person immediate access to commonly faulty components such as the drain pump.



Figure 46. The three tabs for securing the front panel (shown in red)

Finally, the upper two screws, which attach the front panel to the side and rear panels, are replaced with two snap-fit connectors, see figure 47. The front panel is detached in the upper section by pushing in from the sides.



Figure 47. The two upper screws are replaced with snap-fit connectors

The front panel in its detached position with its fasteners highlighted is shown in figure 48.

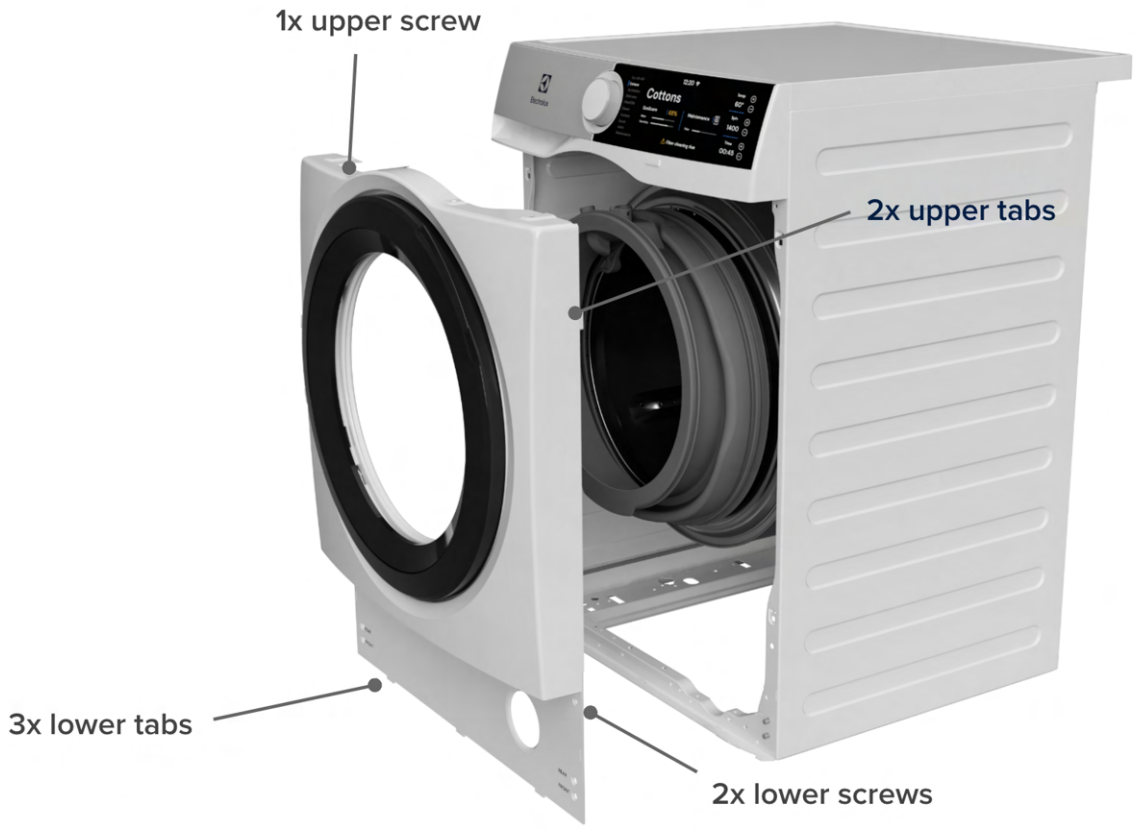


Figure 48. The front panel in its detached position with its fasteners highlighted

7 DISCUSSION

This chapter discusses the process and methods as well as both results, i.e. the PaaS Design Principles and the PaaS Design Proposal

7.1 Process and methods

Using different methods such as observations, various interviews, and product autopsy, gave us a range of insights and a broader understanding that was essential for this project's results. A quantitative research method such as a survey would be beneficial in the future to strengthen or challenge the insights and create a basis for statistical analyses. That could also be used to prioritise the importance of the different PaaS Design Principles so that evaluating the design proposals would become more efficient and accurate. However, the subject of PSS is vast, and to extract the most out of surveys, they should be focused on specific areas of interest, as this project has exhibited.

Furthermore, additional user studies could also have been done to create a weighted evaluation matrix that could evaluate how well the different ideas fulfil the considerations and make it possible to compare them more efficiently. As Van den Berg and Bakker (2015) also mention in their research, having a weighted system is important for accurate evaluations. A future weighted system for the design principles should be developed through further user studies to prioritise insights.

Insights from different stakeholders proved challenging to cluster initially since they covered very different areas of the PSS, and it was not clear how they could be used for product design or correlate in a future PSS business model. It would probably have been more obvious if there had been a developed business model to start with; hence we recommend performing future projects with a larger emphasis on the business and processes of the provider. It can be argued that more insights than needed were acquired in this project, although PSS models cover many areas vital to understanding, and in that regard, they all comply.

Collecting insights regarding consumer experience also gained many insights that could be used for regular sales machines, and not all may have been specific to PaaS. Whilst true that there are overlaps, e.g. making the product easier to repair should also offer a better consumer experience, it can also incur added costs for product development and manufacturing. This could make the product less price competitive and take longer to market, leading to a lower priority in the development process. In a PaaS business model, the provider is economically responsible for the machine, and the incentive will naturally be stronger for the company to add additional features as our principles suggest, hence given a higher priority.

The creative workshop was a method that worked very well to evaluate the principles and generated a lot of interesting ideas and discussions. It showed that people with backgrounds in different fields bring many valuable and various insights. Since PSS systems cover so many

different aspects, it can be hard for one person to consider them all at once. Therefore, having people with backgrounds proved an efficient way to ideate and evaluate different ideas for PaaS.

7.2 PaaS Design Principles

Many design principles can be used for products other than washing machines, which could be interesting to investigate further. Specifically, Ensuring maintenance, Trust and transparency, and Seamless interactions are principles that arguably can be applied to other products, as they are vital for the development of the business model in general and encourage the consumer to perform maintenance and extend products' lifetime. However, as demonstrated in two principles, secondary users such as technicians and installers were important when designing laundry appliances for PaaS, although they may not be equally important for smaller appliances that do not require built-in installations.

Since the consumer will see the people that interact with them through the service as representatives of the provider, it is imperative to have strong partnerships with, e.g., competent installers and technicians. As identified in this project, external factors such as the consumers' drain pipes can affect the machine's performance and require a deeper understanding than just information regarding the machine. To provide an outstanding consumer experience, it is also important that they are positive and polite towards the consumer and that all interactions with the service are delightful, which should be facilitated when the product design adheres to our principles.

During our research, we mainly asked consumers in interviews how accepting they are in terms of cosmetical damages of used products, in attention to their spoken needs. Observations should investigate this further to understand their tacit needs to draw more accurate conclusions regarding their acceptance levels. Understanding this is essential for optimising the PaaS system and ensuring consumers get an outstanding experience.

Creating attractive and available consumer products is vital to make business models more competitive. Our research showed that mainly high-end products with more features and functions are such products in a PaaS system. This implies that there may be more resources needed for production, leading to a higher environmental impact on the environment, although it may be negligible due to the improved lifetime of the product. Meanwhile, making the products more available by offering high-end features at reasonable monthly fees may lead to more products in circulation since it will be easier for people that otherwise would not buy a machine due to large one-off payments to acquire a PaaS machine instead. Therefore, whether the new business model should target consumers that otherwise would have bought a machine should be investigated. Note that this can lead to a risk of the manufacturer cannibalising sales of their conventional machines.

In the context of Sweden, it can be argued that communal laundry rooms already endorse more sustainable consumption. Moreover, making machines more available by a lease model could lead to more people acquiring machines, e.g., to their apartments, even though they already have

access to a washing machine. Hence, balancing a competitive business with sustainability can be challenging, and tools to evaluate the product's potential should be investigated further. From this point of view, it is also interesting to investigate the possibility of utilising our design principles for communal laundry room machines to enable a sustainable service-based business model and expand the target group.

Lastly, the workshop showed that the principles could generate many ideas and discussions, and using them in the future would probably need several ideation sessions; ideally, one for each principle would be more focused and efficient.

7.3 PaaS design proposal

The PaaS design proposal incorporates aspects from all principles; in this regard, our design proposal can be considered complete. However, further validation and testing are needed to determine how well it performs in a real-life situation and meets the users' needs. Ideally, a set of several design proposals would have been preferred to evaluate them against each other. Due to time constraints and such a holistic approach to the product, we did not iterate extensively on each product change. Nonetheless, the changes made are feasible since they use mechanical principles that can be found on existing products which share commonalities with washing machines. For instance, the idea of worm gears for adjusting the feet is implemented on dishwashers currently.

Furthermore, we were limited in the sense that we had to adapt an existing platform. An architectural redesign would have enabled more ideas to flourish. According to Van Den Berg and Bakker (2015), designing for CE involves simplifying the platform architecture. However, this can only be performed to an extent, for example, until the structural rigidity is too compromised. At that stage, it is arguably better to redesign the product from the ground up. An alternative approach would be to start with professional-grade machines with longer engineered lifetimes than consumer appliances. Their components are engineered to exceedingly high operating hours and feature materials that meet tougher requirements. Unfortunately, Electrolux has parted with its professional appliance division, which can be a disadvantage compared to its competitors. Conversely, it can be advantageous because Electrolux can design WMs for PaaS based on an entirely new architecture instead of reusing an old platform.

Additionally, our proposal enables more accessible components from the front. Removing the front panel grants direct access to components such as the drain pump. The on-site repairability is improved since the machines can stay installed during servicing. These changes align with the guidelines devised by Van Den Berg and Bakker (2015) for product architecture and maintenance.

Note that this project has not investigated material choices but is also integral to circular product design. Although recycling is the R-strategy that keeps the least of the product's value in a circular economy (The Ellen MacArthur Foundation, 2015), the recyclability of materials is always important when moving from a linear economy. Our proposal persists with using metal

for the panels, which can be recycled. The issue is more apparent in other areas of the machine, most notably the concrete used as balancing weights for the drum.

Moreover, our PaaS design proposal incorporates additional sensors to perform diagnostics and prognostics better. We have not gone to great lengths to consider these systems' redundancy in case either fails. That said, none of the changes made to the design are safety-critical, i.e. with severe outcomes in the event of failure.

Regarding the UI elements of the machine's interface, it can be argued that disguising the kWh consumption contradicts transparency. However, this should be weighed against the effectiveness of nudging the consumer to be more sustainable. We reason that a bar with an arbitrary scale will likely influence the consumer to adopt sustainable behaviour. This assumption can be tested with a specific research method in future work, e.g. observation. This is also viable for other UI parts, e.g., the maintenance bar. We have attempted to add personality to the machine in line with EDD, aiming to fortify the emotional bond with the product and adopt caring behaviour. We do not claim that this is the ultimate way of achieving it; we have only suggested one alternative which holds potential in this regard. Symbols can be perceived differently by people, and there are cultural aspects to consider in this area, which ought to be examined in future work.

Lastly, our design proposal still relies on a person to physically perform maintenance, e.g. the drain pump filter is required to be cleaned by hand. Exploring other solutions and technologies that possibly remove the need for such maintenance is intriguing. According to Tukker (2004), there may be a risk of inducing careless behaviour when adopting a Product lease model. Hence, it can be argued that it would be safer for the provider to force the user to perform maintenance. However, our research contradicts this risk mentioned by Tukker in terms of consumers being more cautious of products they do not own. While this might be the case in Sweden, it might differ between different cultures and contexts.

8 CONCLUSION

This chapter concludes the project's results by answering the questions defined in the opening chapter.

8.1 Questions to answer

What prerequisites are appropriate for design teams when designing laundry appliances for a PaaS business model?

Our research showed that more actionable and comprehensible principles for the use phase are needed to facilitate work in the early design process. The PaaS Design Principles were successful for early project stages, such as ideation sessions. The participants appreciated the background presented next to the considerations and that they were actionable. With advantage, the principles we developed could be used within the framework by Van den Berg and Bakker (2015) when moving into detailed design.

Although washing machines have been studied explicitly in this master thesis project, several guidelines might also apply to other major domestic appliances. Specific insights for PaaS can also be used for regular consumer sales machines and could be an advantage for both to ensure a low production cost. However, considering the installers and technicians may be specific to built-in installation products.

Research has shown that designing for PaaS requires a multidisciplinary approach; hence the scope needs to involve more stakeholders than when designing for regular sales machines. Therefore, engaging R&D, management and the design teams and ensuring solid processes for cross-department cooperation is needed when designing for PaaS. Some studies also suggest employing new proficiencies specifically for PaaS to facilitate the development.

How can a circular PaaS business model affect how laundry appliances are designed?

The basis for answering this question is that PaaS entails a range of areas that should be addressed. According to our findings, designing laundry appliances for PaaS begins with recognising that the experiences and services surrounding the tangible product should also be considered. Regarding effective repairability, our research indicates that it is important for principles to emphasise the prolonging of the lifetime of products. As mentioned, a longer product life can benefit both the provider and the user. The latter is content with long-term durability while the former retains the value of their asset longer, not to mention the boosted environmental sustainability of preventing functional obsolescence. Thus, our research shows that designing for PaaS entails life cycle thinking and incorporating CE strategies, e.g. refurbish, repair or recycle. In fact, all of the strategies presented on the right-hand side of the Butterfly diagram (n.d.) might come into play when designing for PaaS. However, this project focused on targeting the innermost loop through facilitating on-site repairability specifically. According to

us, this is highly desirable since the products can stay in the hands of the consumers and have the potential to loop from user to user without returning to the service provider or manufacturer.

Another aspect of designing for PaaS and CE is that the product can stay relevant over time. As shown by Van den Berg and Bakker (2015), future-proofing products are important for ensuring that products sustain long-term performance. We have taken steps to ensure that by enabling the product to be upgraded effectively, thanks to OTA updates. Also, the removable front panel can be replaced for cosmetic reasons if that is needed.

Moreover, our research suggests that designing for PSS requires a user-centric approach, ensuring that the product and service work in symbiosis to deliver an outstanding user experience. This necessitates a holistic perspective and an understanding of the flow of the product's journey in the PaaS model, recognising categories of users, such as installers and technicians who also interact with the consumers.

Finally, PaaS models differ from conventional sales in shifting from pure products to providing services. Laundry appliances could be prepared for various business models, such as Pay-per-use, by integrating them into the design through connectivity, incorporating advanced sensors or other means.

8.2 Recommendations for future work

- Investigate whether people treat different products differently (EDD) and if they treat WM worse because of the leasing model, preferably through research methods as e.g., observations to minimise the Hawthorne effect and researcher bias
- More thorough investigations of consumers acceptance in terms of cosmetical damages and machine freshness
- Future cost calculations are recommended for the suggested design proposal and the overall PSS business model
- Cost optimisation is important and advised, as mentioned by Loon, Van Wassenhove and Mihelic (2021)
- The use of EDD in facilitating maintenance may be intriguing and should be researched more in-depth
- Evaluation matrix could be weighted based on more user insights for better precision, alternatively use the Spider map by Van den Berg and Bakker (2015)
- Explore possibilities to apply the principles to further product categories
- Fitting wheels to the machines is interesting and can be investigated further
- Investigate polishing of the drum's outside to prevent that mould and bacteria deposits stick to the drum or even new materials
- Perceptual mapping can be a beneficial to understand target groups or choice of brand position
- Explore alternative technical solutions that eliminates the need for physical maintenance to minimise the risk of product obsolesence

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APPENDIX A: CE DESIGN GUIDELINES

Circular design guidelines proposed by researchers from TU Delft (Van den Berg and Bakker, 2015).

Category	Sub-category	Goal	Means	Source
Futureproof last long and use long	Long lasting	Performance		
		Reliability	Design out moving parts	(Mulder, et al. 2014)
			Design for under stressed use	(Mulder, et al. 2014)
			Provide redundancy	(Mulder, et al. 2014)
			Over dimension critical components	(Mulder, et al. 2014)
		Durability	Wear resistance	(Sundin, 2004)
			Use assembly methods that allow disassembly without damage to (reusable) components.	(Ijomah, et al., 2010)
			Do not use coated, painted or plated components	(Mulder, et al. 2014)
			Prevent discolouring	
			Ensure that fasteners' material are similar or compatible to that of base material thus limiting opportunity of damage to parts during disassembly.	(Ijomah, et al., 2010)
			Aging and corrosive material combinations need to be avoided, since disassembling them cleanly and efficiently (due to their tendency to corrode, spread corrosion, and break off inside the product) often is difficult.	(Mital, et al., 2008)
			Protect subassemblies from corrosion, the reasons being the same	(Mital, et al., 2008)
	Long in use	Roadmap fit	Ensure a long -term roadmap is available	
		Upgradability	Use materials and assembly methods that do not prevent upgrade and rebuilding of the product.	(Ijomah, et al., 2010)
			Structure to facilitate ease of upgrade of product.	

		Adaptability	Ensure a long -term roadmap is available	(Ijomah, et al., 2010)			
			Prevent product obsolescence (user needs)	van den Berg			
		Timeless design	Emotional durable design (user desire)				
		Anticipate legislation	(e.g. toxicity, recyclability, disassembly time)				
Disassembly non-destructive	Connections	Quick and easy disconnect	Use easy to disassemble connections				
			Apply loose fits for internal components	(Peeters, et al., 2012)			
			Avoid welding and adhesive between sub-assemblies	(Ijomah, et al., 2010)			
			Use joining methods that allow disassembly at least to the point that internal components and subsystems requiring it can be accessed for testing before and after refurbishment.	(Ijomah, et al., 2010)			
		Limit use and diversity of fasteners	Minimize the number of fasteners used in an assembly	(Mital, et al., 2008)	(Peeters, et al., 2012)	(Ijomah, et al., 2010)	(Balkenende, et al., 2011)
			Minimize the types of fasteners used in an assembly and standardize the fasteners used	(Peeters, et al., 2012)	(Ijomah, et al., 2010)	(Balkenende, et al., 2011)	
			Fasteners need to be easy to remove or destroy.	(Mital, et al., 2008)			
			Allow easy access and identification of the fasteners	(Mital, et al., 2008)	(Sundin, 2004)	(Ijomah, et al., 2010)	(Balkenende, et al., 2011)
			Consider the use of fasteners incorporating an active disassembly or embedded disassembly functionality.	(Balkenende, et al., 2011)			
		Limit use and diversity of tools	Limit the number of tools needed and tool changes	(Balkenende, et al., 2011)			
			Make it possible to use simple tools for disassembly	(Balkenende, et al., 2011)			
	Product architecture	Simplify product architecture	Minimize the complexity of the product structure	(Desai & Mital, 2003)	(Ijomah, et al., 2010)		
			Select a product structure which allows a sequence independent disassembly	(Balkenende, et al., 2011)			
			Minimize the number of components used in an assembly	(Mital, et al., 2008)	(Desai & Mital, 2003)	(Ijomah, et al., 2010)	(Balkenende, et al., 2011)

			Optimizing the spatial alignment between various components to facilitate disassembly without jeopardizing assemblability,	(Desai & Mital, 2003)		
			At least one surface needs to be left available for grasping.	(Mital, et al., 2008)	(Sundin, 2004)	
			Simplify and standardize component fits	(Ijomah, et al., 2010)		
		Ease of access to components	Arrange components for ease of disassembly	(Ijomah, et al., 2010)	(Balkenende, et al., 2011)	
			Consider making the plane of access to components the same for all components	(Mital, et al., 2008)	(Sundin, 2004)	(Balkenende, et al., 2011)
			Avoid the need to turn the product in the disassembly process	(Mital, et al., 2008)	(Sundin, 2004)	(Balkenende, et al., 2011)
			Metal inserts in plastic parts should be avoided, since this increases material variety and part complexity and necessitates multiple directions and complex movements in disassembly. Applicable if meant for over moulding	(Mital, et al., 2008)		
			Use assembly methods that would allow disassembly at least to the point that internal components and subsystems requiring work can be accessed.	(Ijomah, et al., 2010)		
		Clarity of disassembly sequence	Identify components assembly sequence.	(Ijomah, et al., 2010)	(Sundin, 2004)	
			Identify components requiring similar assembly tools and techniques.	(Ijomah, et al., 2010)	(Sundin, 2004)	
			Reduce complexity of reassembly	(Ijomah, et al., 2010)		
Maintenanc	Maintenance	Ease of cleaning	Ensure product surfaces are smooth and wear resistant.	(Ijomah, et al., 2010)		
			Ensure that all parts to be cleaned are easily accessed.	(Ijomah, et al., 2010)	(Peeters, et al., 2012)	(Sundin, 2004)
			Use material that would survive cleaning process e.g. ensure that material melting	(Ijomah, et al., 2010)		

	e Reuse of products			point is higher than clean process temperature.		
non-destructive				Limit the number of material types per part.	(Ijomah, et al., 2010)	
				Identify components requiring similar cleaning procedures and cleaning agents.	(Ijomah, et al., 2010)	(Sundin, 2004)
			Ease of repair	Allow for easy and quick access to parts prone to failure	(Peeters, et al., 2012)	(Ijomah, et al., 2010) (Sundin, 2004)
				Avoid assembling components with a different life duration	(Balkenende, et al., 2011)	
			Allow onsite repair and upgrade	Allow on-site maintenance		
		Lifetime prognostics	(Online) monitoring for quality, testing, maintenance and billing			
	Remake Reuse of components	Modularity	Use modular components	Use modular structure so that obsolescence occurs with components rather than with entire product.	(Ijomah, et al., 2010)	(Mital, et al., 2008) (Ijomah, et al., 2010)
				Do not combine components that have different physical life.	(Hata, et al., 2001)	
				Do not combine components that have different intervals for maintenance and upgrade.	(Hata, et al., 2001)	
				Group components in sub-assemblies according to reuse, reconditioning or remanufacturing potential	(Balkenende, et al., 2011)	
				Concentrate compatible material groups in separate subassemblies of a product	(Mital, et al., 2008)	(Balkenende, et al., 2011) (Hata, et al., 2001)
				Allow customization by grouping components in liberally	(Balkenende, et al., 2011)	
				Combinable subassemblies	(Balkenende, et al., 2011)	
			Use standard interfaces	Standardize parts	(Ijomah, et al., 2010)	
				Standardize interfaces	Maarten	

			Back- & forwards compatibility		Nestor Palma		
		Reliability assessment	Allow for easy testing of components	Standardize test procedures	(Ijomah, et al., 2010)		
				Structure for ease in determining component condition	(Ijomah, et al., 2010)	(Sundin, 2004)	
				Structure so testing is sequential, mirroring reassembly order	(Ijomah, et al., 2010)		
				Minimize the disassembly level required to effectively test components	(Ijomah, et al., 2010)		
				Clearly identify component load limits, tolerances and adjustments	(Ijomah, et al., 2010)	(Sundin, 2004)	
		(Reverse) logistics	Product can easily be returned	Ensure products can be stacked			
				Ensure products can safely be transported			
				Minimize product volume			
			Allow for spare part harvesting				
			Local production				
non-destructive & destructive	**Recycling** Reuse of material	Materials	Avoid the use of (non-compliant) coatings	Any secondary coating processes, such as painting, are to be avoided, since they inhibit access to and removal of components	(Balkenende, et al., 2011)	(Mital, et al., 2008)	(Hultgren, 2012)
			Limit the number of different materials	Minimize the number of material types used in an assembly	(Balkenende, et al., 2011)	(Ijomah, et al., 2010)	(Hultgren, 2012)
			Only use recyclable materials		(Balkenende, et al., 2011)	(Hultgren, 2012)	
			Use preferred/pure materials	Increase the use of common materials	(Balkenende, et al., 2011)	(Desai & Mital, 2003)	
			Allow material separability	Consider the material compatibilities to eliminate the need of separation for recycling	(Balkenende, et al., 2011)		
				Allow easy material identification	(Balkenende, et al., 2011)	(Mital, et al., 2008)	(Sundin, 2004)
				Add non-contamination markings for the ease of sorting and recycling	(Balkenende, et al., 2011)	(Mital, et al., 2008)	

				Any harmful materials, if functionally important, should be grouped together into subassemblies for fast disposal.	(Balkenende, et al., 2011)	(Mital, et al., 2008)
				Do not use fasteners that are not compatible with the connecting components. Fasteners are recycled together with the host component; therefore choose plastic fasteners for plastic and metal fasteners for metal to avoid polluting other material streams or end up in the waste fraction	(Hultgren, 2012)	
		Electronics	Get PCB out in one piece		(Balkenende, et al., 2011)	
			Easy/fast detection of materials		(Balkenende, et al., 2011)	
			Use SMD components		(Balkenende, et al., 2011)	
		Connections	Avoid fixed connections	Prefer snap-fits for plastic components (particularly housing), to allow easy liberation of materials	(Balkenende, et al., 2011)	(Hultgren, 2012)
				Use a detachable power cord instead of a permanently fixed one	(Hultgren, 2012)	
			Break down by (shredding/disassembly) to	If connections are applied that enclose materials permanently, apply gaps and or break-lines to the enclosing material to enable liberation during shredding	(Balkenende, et al., 2011)	(Hultgren, 2012)
			Pieces of uniform composition		(Balkenende, et al., 2011)	
			Pieces of relatively large size (>1cm)		(Balkenende, et al., 2011)	

APPENDIX B: INTERVIEWS WITH ELECTROLUX

Organisation

- Tell us a bit about yourselves and your roles at Electrolux
- How did you come into this role?
- What is your work about, and what are your purposes?
- How is your team working today, what processes are being used?
- How do you cooperate with other departments?
- What are you working on currently?
- What difficulties is your department facing today?

Design

- How are the design team working today, what processes are being used?
- Differences between different design teams?
- What difficulties are the design teams facing today?
- How are design guidelines being followed today? (or not followed)
- How does your work affect the product design?
- For WMs specifically, what is important for your department?

General

- How would you and your department approach our task?
- What stakeholders do you think are essential to include?
- Do you know of any other projects that may be of interest to our project?
- In what way do you think that our project can contribute to your work?
- Do you see any trends that are relevant to our project?
- In what way do you think that our project can contribute to your work?
- How would you approach our task?

APPENDIX C: INTERVIEWS WITH PRIMARY USERS

Demographics

- Namn
- Kön Man / Kvinna
- Hur gammal är du?
- Har du studerat något eftergymnasialt?
- Student/ej student
- Vad jobbar du med?
- Vart bor du?
- I vilken typ av bostad bor du i?
    - Storlek
    - Antal rum
- Äger du bostaden eller hyr du den?
    - Hyres i 2:a hand / 1:a hand?
- Bor du själv eller med en partner/familj? (hur många i hushållet?)

Personality

- Hur skulle dina vänner beskriva dig som person?
- Vad brukar du göra på din fritid?

About washing machines

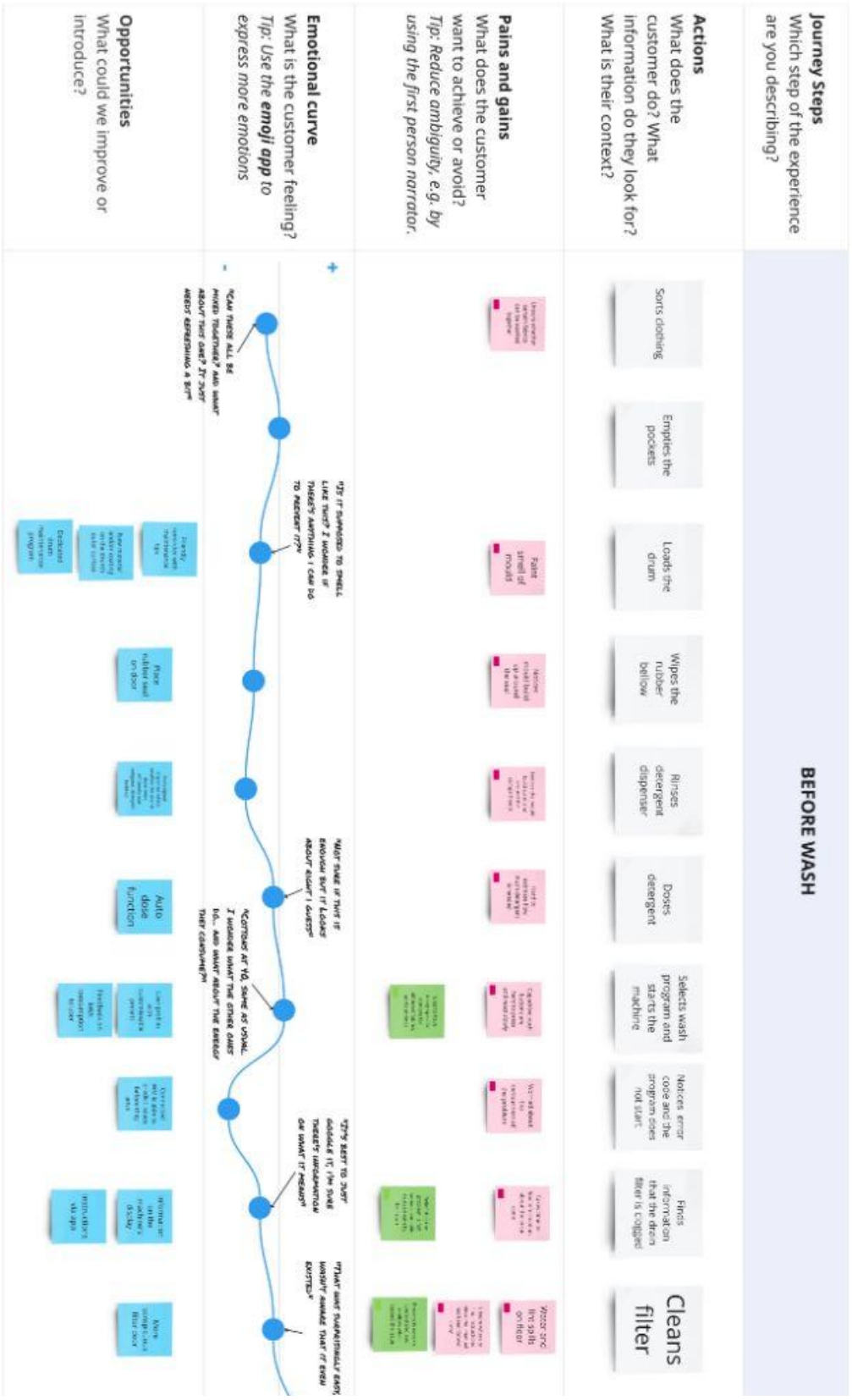
- Har du tvättmaskin i din bostad och/eller i tvättstuga?
    - **Om båda:** Var tvättar du & varför där?
- Har du valt/köpt maskinen själv?
    - **Om köpt:**
    - Begagnad / ny? Varför?
    - Vilken modell valde du och varför?
    - Hur gammal är den?
    - Vad tycker du om din tvättmaskin?
    - Vilka egenskaper är viktiga för dig vid anskaffandet av en tvättmaskin?
    - Har du installerat en tvättmaskin på egen hand?
    - Vid gångerna du valt att byta tvättmaskin, av vilka anledningar har du gjort det?
    - **Om EJ köpt**
    - Vad har ni för maskin i tvättstugan/badrummet?
    - Vad tycker du om maskinerna i er tvättstuga/den som ingått i bostaden?
    - Vad hade du ändrat om du köpt en maskin själv?
- Hur ofta tvättar du?
    - Stora / små tvättar?

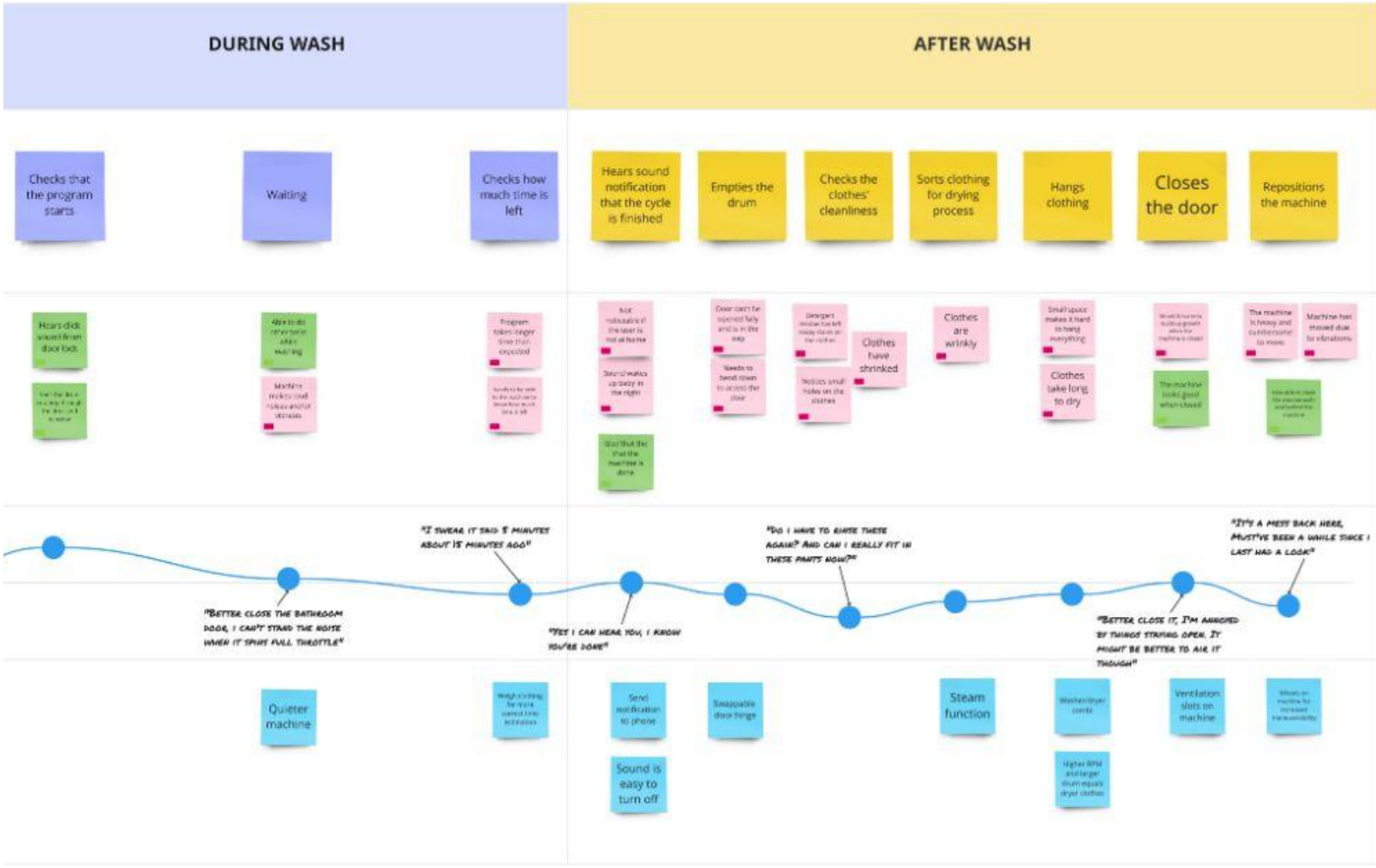
- Vad använder du för typ av tvättmedel?
    - Varför just det?
    - Hur brukar du dosera?
- Hur lärde du dig att tvätta?
- Tycker du att det är enkelt att använda tvättmaskiner?
    - Vad / vad inte? Varför?
    - Upplever du något som svårt?
- Hur brukar du ta hand om tvättmaskiner?
    - Rengöring? Av vad?
    - Känner du att du har tillräckligt med kunskap om hur maskinen ska underhållas?
    - Finns det någon elektronikprodukt du äger som du tar hand om mer än andra? Varför?
- Vad skulle du göra om tvättmaskinen går sönder?
- Brukar du läsa manualen till tvättmaskiner?
    - Om ja, är det oftare på andra produkttyper? Vilka?

Leasing

- Har du någonsin köpt en begagnad vitvara?
    - Varför inte ny?
    - Vilken/vad var viktigt vi köp?
    - Vilka kompromisser var du tvungen att göra?
- Har du hyrt/leasat någon/några produkter?
    - Varför just den produkten?
    - Varför leasa istället för köpa? Nöjd/ej?
- Hur brukar du förhålla dig till produkter du hyr kontra äger?
- Skulle kunna tänka dig att hyra en tvättmaskin istället för att köpa en tvättmaskin?
- Kan du beskriva steg för steg din tvättprocess?
- Finns det något du vill tillägga?

APPENDIX D: CUSTOMER JOURNEY MAP




DURING WASH
AFTER WASH
Checks that the program starts
Waiting
Checks how much time is left
Hears sound notification that the cycle is finished
Empties the drum
Checks the clothes' cleanliness
Sorts clothing for drying process
Hangs clothing
Closes the door
Repositions the machine
Hears click sound from door lock
Clothes have shrinked
Clothes are wrinkly
Clothes take long to dry
"I SWEAR IT SAID 5 MINUTES ABOUT 15 MINUTES AGO"
"BETTER CLOSE THE BATHROOM DOOR, I CAN'T STAND THE NOISE WHEN IT SPINS FULL THROTTLE"
"YES I CAN HEAR YOU, I KNOW YOU'RE DONE"
"DO I HAVE TO RINSE THESE AGAIN? AND CAN I REALLY FIT IN THESE PANTS NOW?"
"BETTER CLOSE IT, I'M ANNOYED BY THINGS STAYING OPEN. IT MIGHT BE BETTER TO AIR IT THOUGH"
"IT'S A MESS BACK HERE, MUST'VE BEEN A WHILE SINCE I LAST HAD A LOOK"
Quieter machine
Send notification to phone
Swappable door hinge
Steam function
Ventilation slots on machine
Sound is easy to turn off